MEC-102

MACROECONOMIC ANALYSIS

School of Social Sciences
Indira Gandhi National Open University
Maidan Garhi, New Delhi-110068

EXPERT COMMITTEE

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Unit 19	Fiscal Sustainability	

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COURSE INTRODUCTION

Macroeconomics analyses and establishes functional relationships among economy level aggregates. Aggregative analysis has assumed such a great significance in recent times that a prior understanding of macroeconomic theoretical structure is considered essential for proper comprehension of various issues and policies. Macroeconomics now is not only a scientific method of analysis but also a body of empirical economic knowledge.

After going through this course a student is expected to

- have an intuitive understanding of the theoretical developments in macroeconomics;
- contrast among various schools of macroeconomic thought;
- analyse real world issues with the help of macroeconomic theories; and
- apply macroeconomic theory to empirical data.

The course comprises seven blocks.

Block 1, entitled **Traditional Approaches to Macroeconomics**, is somewhat introductory in nature. After explaining some basic concepts that are used frequently in macroeconomics and form building blocks of the subject, it presents the classical and Keynesian views on determination of output and employment in an economy. Subsequently, it presents a synthesis of the classical and Keynesian ideas through the tools of IS-LM curves. The block also deals with issues pertaining to open economy macroeconomics.

Block 2, entitled **Expectations and Macroeconomics**, deals with issues such as inflation and unemployment. It traces the evolution of Phillips Curve – how it originated as a relationship between real and monetary variables; how it occupied a prime place as an analytical tool in Keynesian economics; and how it became irrelevant in the new-classical economics. In the course of discussion, this block deals with two important concepts, viz., adaptive expectations and rational expectations.

Block 3, entitled **Intertemporal Decision-making**, analyses the behavior of economic agents such as households and firms in an inter-temporal framework. It discusses the Ramsey-Cass-Koopmans model and overlapping-generations model. It also looks into issues such as the choice between work and leisure on the part of households.

Block 4, entitled **Theories of Business Cycles**, is devoted to the analysis of economic fluctuations in an economy, particularly the business cycles. It begins with traditional theories of business cycles and later moves on to real business cycles models with inter-temporal substitution and the propagation mechanism.

Block 5, entitled **Labour Markets**, discusses issues related to unemployment in the economy. Apart from presenting nominal and real rigidities, the block throws lights on search theories and unemployment.

The subject matter of **Block 6** is **Issues in Monetary Policy**. It deals with issues such as money supply, conduct of monetary policy, and theory of monetary policy. In recent years inflation targeting, policy rules, and dynamic consistency have been important concerns of policy makers.

Block 7, entitled **Fiscal Policy and Macroeconomics**, looks into the implications fiscal policy in an economy. Debt sustainability and Ricardian equivalence are two important issues discussed in this Block.

UNIT 1 THE CLASSICAL APPROACH*

Structure

1.1 OBJECTIVES

After going through the unit, you should be in a position to

- bring out the salient features of classical economics;
- analyse the classical approach to the determination output and employment;
- explain the quantity theory of money and its implication on price determination;
- explain the classical dichotomy in the economy and the neutrality of money in a classical system; and
- describe the behavior of prices in the long-run and short-run.

1.1 INTRODUCTION

Being a student of economics, you must be aware that macroeconomic theory has evolved over time in response to the dynamic macroeconomic environment. Initially, during the nineteenth and early twentieth century, there was a consensus among economists and economic historians that an economy could run smoothly without much fluctuation in income, output and employment. The perception

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was that there should be minimum state intervention in economic variables such as wages, prices and interest rates. The market forces (supply and demand) would take care of economic equilibrium in an economy. Such an assumption, however, is unrealistic – all governments have in place certain economic policies (such as monetary policy and fiscal policy) and they carry out amendments to these policies from time to time!

Economic theory refers to a set of ideas and principles which are used to explain, analyse and predict certain 'phenomenon' or observable events. As you know, macroeconomic theory deals with macroeconomic phenomena, which are aggregative in nature. The need for a special branch of macroeconomics arises because what holds for the individual units may not hold good for the economy as a whole. For example, suppose a firm employs labour for production of output (say, cement). It can hire as many workers it requires at the ongoing wage rate. Thus, increase in demand for labour by a single firm does not have any impact on the wage rate. However, if all the firms in a country increase their demand for labour (say due to economic boom and optimism in the country), there will be shortage of labour and increase in wage rate. Further, the number of workers available for work in the country is limited; thus demand for labour beyond this limit will increase wage rate only, not the supply of labour.

Let us look into the global energy crisis of 2022. This took the form of global shortages and increased the prices of oil, gas and electricity throughout the world. The crisis was caused by a variety of social and economic factors: labour shortages, political disputes, climate concerns, and the war between Russia and Ukraine. The world economy responds to such crises in certain ways. During the global energy crisis of 2022, for example, the oil-exporting countries witnessed an increased inflow of income while petroleum-importing countries experienced a rise in the cost of production. There was severe inflation in most countries. In order to control inflation, policy makers resorted to increase in interest rates and reduction in money supply.

Countries design their monetary and fiscal policies for the welfare of people by controlling inflation and boosting economic growth. Macroeconomic theory guides economists and policymakers to explain the situation and design appropriate policy measures. Among economists, however, there is no agreement on how equilibrium levels of output, prices and employment are determined. Historically economic data are the same, but economists differ on the reasons and solution for economic problems. In microeconomics there is more or less some consensus. In macroeconomics, as you will see, there are sharp and often altogether different explanations and policy recommendations for the same phenomenon.

Macroeconomics has been subjected to debates, particularly since the 1930s. The reasons behind such differences among economists are the basic assumptions and the models they take. Two major schools of thought – Classical and Keynesian – have interpreted economic events differently. Accordingly, they have accorded different explanations for the same economic issue. In this unit, we focus on the classical theories while Keynesian theories will be discussed in Unit 2.

1.2 VARIOUS SCHOOLS OF MACROECONOMIC THOUGHT

A school of macroeconomic thought refers to a group of macroeconomic thinkers who share similar views on the behaviour of macroeconomic variables in an economy. As pointed out above, economists do not always share a similar approach in dealing with macroeconomic problems. For example, Keynesian theory came into existence as a response to the Great Depression. Further economic events during the 1970s led to the revival of the classical views in the form the New Classical Economics. As a response to New Classical economics, a new school of thought – New Keynesian Economics – was born by introducing micro-foundations and rational expectations in Keynesian economics. In later Units of this we will discuss about both these schools of thought.

Classical economics refers to a school of thought prevalent during the eighteenth and early nineteenth century. They believed that the levels of economic variables in an economy were determined through market forces. It implies that wages, prices and interest rates are flexible in an economy. However, the occurrence of the Great Depression (1929-1933), which witnessed massive decline in income, output, employment and price level, a stock market crash, and banking panic falsified such assumptions. This led to the search for new explanations periodic fluctuations in output and employment in an economy. John Maynard Keynes, a British economist, opposed the classical ideas and advocated that prices and wages are not flexible; rather they are sticky. He coined the term “classical” to reflect the ideas presented by economists prior to him. Prominent among classical economists are Adam Smith, David Ricardo, Thomas Malthus, Anne Robert Jacques Turgot, John Stuart Mill, Jean-Baptiste Say, and Eugen Böhm von Bawerk.

As opposed to the classical economists, Keynes advocated that the government should actively intervene in the market to counter recessionary conditions. From Keynes’s beliefs evolved a new school of macroeconomic thought that is called the Keynesian school. According to Keynesian economists, actual output could be lower than potential output of an economy and the gap can be closed through fiscal and monetary policy interventions.

There are two major schools of thought: classical and Keynesian. Classical theory emerged in the 18^{th} and 19^{th} centuries. It attempted to explain and analyse the remarkable increase in wealth. Classical theory relies on the assumption of fully flexible prices and wages, which results in an automatic adjustment in the market, ensuring no deficiency of aggregate demand. The economy operates at its full capacity or full employment level. These classical ideas failed to explain and understand the great depression of the 1930s. The depression had affected all the major advanced countries, wherein about one fourth of the labour force had become unemployed. Against this backdrop, John Maynard Keynes came up with his ideas, called Keynesian theory. He attempted to explain the event and advocated that such huge unemployment was because of inadequate aggregate demand, which in turn was the result of a deficiency in investment demand. He criticized the assumption of a *laissez-faire* economy in the classical theory and advocated policy intervention to increase demand and employment.

Later, attempts were made to integrate ideas from both the schools, which led to the “neoclassical synthesis”. The IS-LM model that we discuss in Unit 3 is an outcome of the neoclassical synthesis. During the 1950s, economists such as Harrod, Domar and Kaldor extended the Keynesian ideas from short-run to long-run. They argued that there would be an increase in production capacity in the long run because of large investments. This in turn would boost economic growth. The focus of these economists, called the Post-Keynesians, was to understand, explain and analyse economic growth. In order to overcome certain limitations of the post-Keynesian growth models, Robert M. Solow came up with the neoclassical growth model. You will go through these growth models in the course, ‘Economics of Growth and Development’.

Keynesian school of thought dominated economic theory and policy after the World War II until the 1970s. During the 1970s most of the advanced economies witnessed stagnation and inflation, a situation called “stagflation”. Keynesian theories did not have appropriate policy response to stagflation, which led to the exploration of fresh explanations of economic behavior. Robert E. Lucas, an American economist, revived the classical ideas and countered the prevailing Keynesian thought. The ideas of classical and Keynesian schools differ from each other on various issues, including: i) the role played by demand and supply in determination of equilibrium output, employment and prices; ii) the assumption of flexibility of price level and wage rate in the economy; iii) the dichotomy of real and monetary sectors of the economy, and (iv) neutrality of money.

After the 1970s, however, there was revival of classical economics in the form of new-classical theory. In response to that, new-Keynesian models came up. New Classical economics retained the basic characteristics of classical economics but introduced two new concepts: rational expectations and micro-foundations. We will cover these aspects in later units of this course.

Along with these mainstream schools of thought, there is heterodox schools of thought or heterodox economics. It includes Marxist, Austrian, institutional, feminist and social economics among many others. In fact, there is a need for promoting pluralism in economics these days and researchers are venturing into many new areas. The mainstream or conventional economics, as mentioned, earlier have been Classical and Keynesian. The legacy of these two schools of thought have continued in different forms. Two prominent schools of thought in recent years have been New-Classical economics and New-Keynesian economics. There are certain common features of these two schools of thought: (i) general equilibrium models, (ii) micro-foundations in behaviour of economic agents, and (iii) rational expectations. Two major differences between both the schools of thought are: (i) the assumption regarding market structure – the new-classical school assumes that there is perfect competition in the market while the new-Keynesian school assumes market imperfection. (ii) the assumption of flexibility in wages and prices – the new-classical school assumes flexibility in wages and prices while the new-Keynesin school assumes rigidities in wages and prices.

1.3 BASIC FEATURES OF CLASSICAL THEORY

Classical economics evolved against the perception of 'mercantilism'. The mercantilists believed that the wealth of nations depended on the stock of bullion (or, gold and silver) and adopted policies that promoted exports and discouraged imports through subsidies and tariffs. Classical economists discarded such ideas and believed that 'wealth of nations' depended on real factors. For them, money is merely a medium of exchange. Important features of classical theory are as follows:

(i) **Microeconomic Issues**: Classical economists dealt mostly with microeconomic issues pertaining to behavior of economic agents such as firms and households. In classical view, a firm maximizes its profits subject to a resource constraint. Similarly, households try to maximize their utilities or economic gains given their budget constraints. Classical economists believed in the optimising tendencies of the market mechanism. According to classical economists, variables such as price level, wage rate and output level should be determined by market forces (supply and demand).

(ii) **Laissez Faire**: Classical economists believed in the philosophy of 'laissez faire', which is a French term meaning 'leave alone' or 'let you do'. According to this view, there should be minimal intervention from the government in business affairs. In fact, Adam Smith suggested that government should confine itself to three main duties, viz., (i) national defense, (ii) administration of justice (law and order), and (iii) establishing and maintaining certain public works (infrastructure, education, etc.).

(iii) **Invisible Hand**: Adam Smith introduced the concept of the 'invisible hand'. According to him, the economy will function well if everyone pursues his/ her own interest. According to him, "It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own interest". Individuals pursuing self-interest seem to be led by an 'invisible hand' to maximise the general welfare of everyone in the economy. It is not the generosity of a producer in selling a commodity; it is his/her self interest. Similarly, the consumers are not doing a favour to the producer; they are pursuing their own interest in buying the commodity. The philosophy of the invisible hand confined classical economists to the analysis of the behavior of economic agents; they failed to see any conflict between interest of economic agents and that of the economy as a whole.

(iv) **Continuous Market Clearing**: Classical economists assumed that prices and wage rates are flexible. As you know from microeconomics, equilibrium price is determined at the level where supply and demand are equal. Given the supply and demand curves, if demand is more compared to supply, price will increase. Similarly, if supply is more compared to demand, price will decrease. This principle applies not only to commodities, but also to wage rate. If supply of labour is more than its demand, there will be a decline in wage rate till supply of labour equates its demand. An implication of the above is that there is no unemployment in the economy – wage rate will decline till all workers are employed. There is no disequilibrium in the market.

(v) **Perfect Competition**: Classical economists assumed that there is perfect competition in the market so that markets function smoothly. As there is full employment (due to flexibility in wage rate), production is always at the full employment level. An implication of the above is that there is no scope for ups and downs in the level of output. Going by this logic, the classical economists ruled out the possibility of 'business cycle'.

(vi) **Say's Law of Market**: Classical economists believed that production or supply is the key to economic prosperity. Thus, they emphasized more on the supply side of the economy. This approach is summarized very well in the Say's law, named after the prominent classical economist J B Say. According to J B Say, 'supply creates its own demand'. Whenever some production takes place, there is a stream of income in the hands of people, which generates demand.

A person should have produced something to sell (say, for example, labour) and thereby earned certain income. Thus, in the classical viewpoint, there is no scope of deficiency of demand in the economy. Therefore, primary concern before an economy is production or supply, not demand.

(vii) **Neutrality of Money**: According to classical economists, economic growth of an economy is due to increase in the factors of production and technological progress. Money is just a medium of exchange; and it facilitates transactions among economic agents. Thus, increase in money supply does not affect the level of output – it only leads to increase in prices (see quantity theory of money discussed below). They believed that there is dichotomy between monetary variables such as money supply and prices, and real variables such as output and employment. Thus, classical economists stressed the role of real factors in deciding the real variables such as output and employment. As mentioned earlier, classical economists believed in free market systems without government intervention. Determination of output, employment and interest rate in the classical system is given below.

We discuss some of the major issues in the next section.

Check Your Progress 1

1) Write down the important features of classical theory.

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2) What are the main features of new-classical economics?

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1.4 DETERMINATION OF OUTPUT AND EMPLOYMENT

The classical theory is based on the following assumptions:

a) firms and workers are optimisers,

b) they have perfect knowledge, and

c) they operate in competitive (perfect) markets.

An implication of the above is that wages and prices are fully flexible. Under perfect competition, the demand for labour for the profit maximising firms is

$$P = W/MPN \qquad \ldots (1.1)$$

where, P is the product price,

W is nominal wages, and

MPN is the marginal product of labour.

As there is perfect competition, P is equal to Marginal Revenue (MR) which is received from the sale of one unit of output. Here, W/MPN represents the marginal cost of production, that is, the cost of production of an additional unit of output.

Equation (1.1) can be rewritten as

$$MPN = W/P \qquad \ldots (1.2)$$

(We denote nominal wage by W and real wage by its lower case, i.e., *w*)

Equation (1.2) indicates that the marginal product of labour is equal to the real wage (W/P). Hence, the labour demand curve in terms of real wage is nothing but the Marginal Product of Labour. The labour demand curve is downward sloping.

$$MPN_d = f(W / P) \qquad \ldots (1.3)$$

The of supply of labour varies positively with the real wage. It assumes that individual labour maximises its utility.

This can be written as

$$N_S = g(W / P) \qquad \ldots (1.4)$$

where N_S indicates the supply of labour. We explain below in Fig. 1.1 the equilibrium in the labour market. Note that the labour market is in equilibrium at full employment of labour. At the equilibrium point, the supply of labour is equal to demand for labour. In the upper panel, we provide labour supply and labour demand as functions of real wage $\left(\frac{W}{P}\right)$. In the lower panel, we provide labour supply and labour demand as functions of nominal wage. As price level increases from P_1 to P_2 and then to P_3, real wage can be maintained at the same level if nominal wage increases to 2W and 3W respectively. The shifts in the supply curve of labour and demand curve of labour are shown in panel (b) of Fig. 1.1. Notice that there is no change in labour supply, as full employment is ensured (since there is no change in real wage). If there is an increase in output prices, but no increase in nominal wage rate, there will be a decline in real wage. This will lead to a decrease in labour supply.

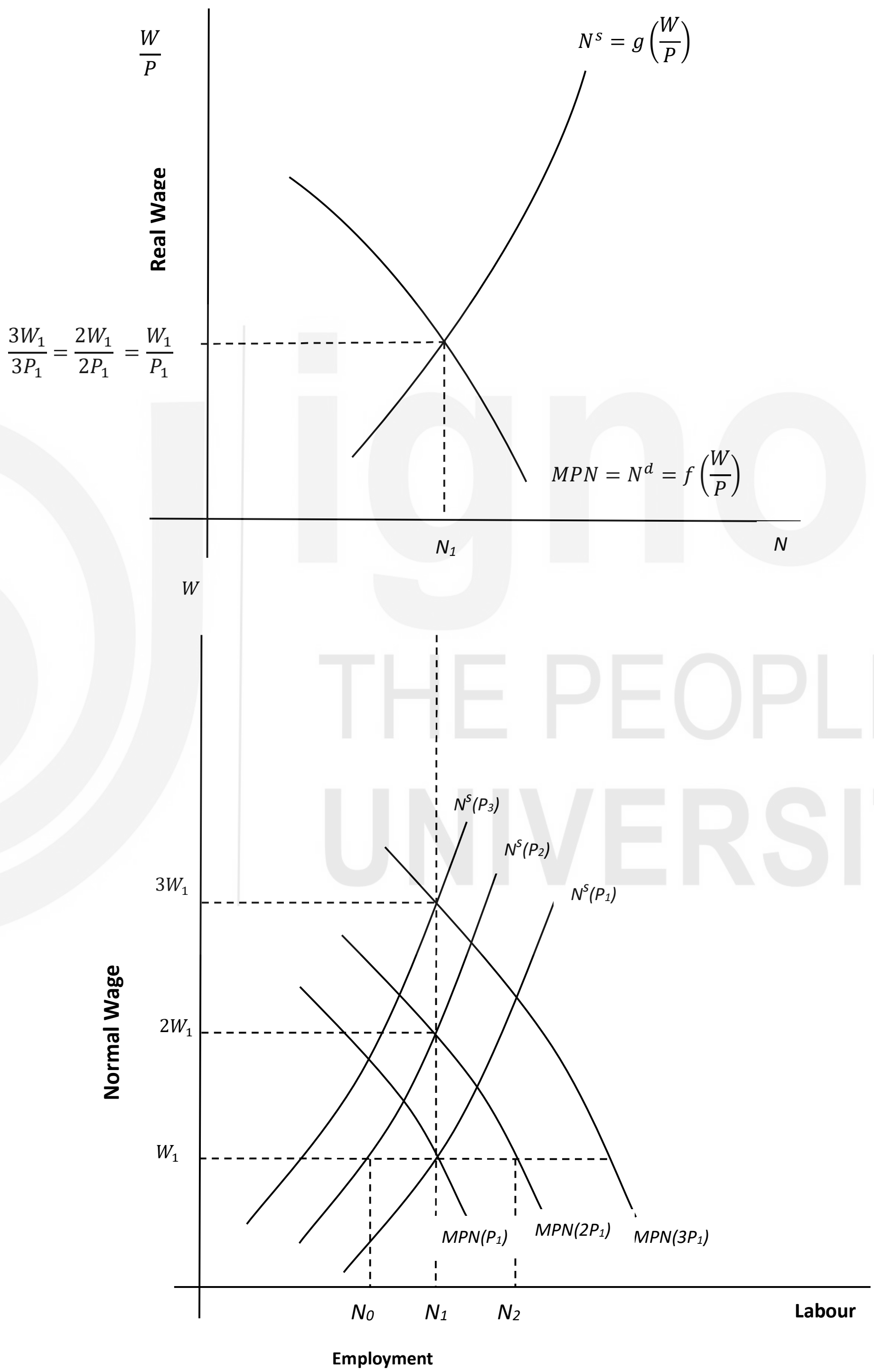

Fig.1.1: Supply of and Demand for Labour

Let us assume that there are only two factors of production, i.e., labour and capital. Output is determined by the production function Y=F(K,N) as given in Fig. 1.2, panel (b). Equilibrium in the labour market (see panel (a) of Fig. 1.2) determines the full employment labour supply and real wage rate. Based on that full employment out is given in panel (b) of Fig. 1.2. As you can see, the economy operates with full employment output (Y_0) and full employment labour supply (N_0).

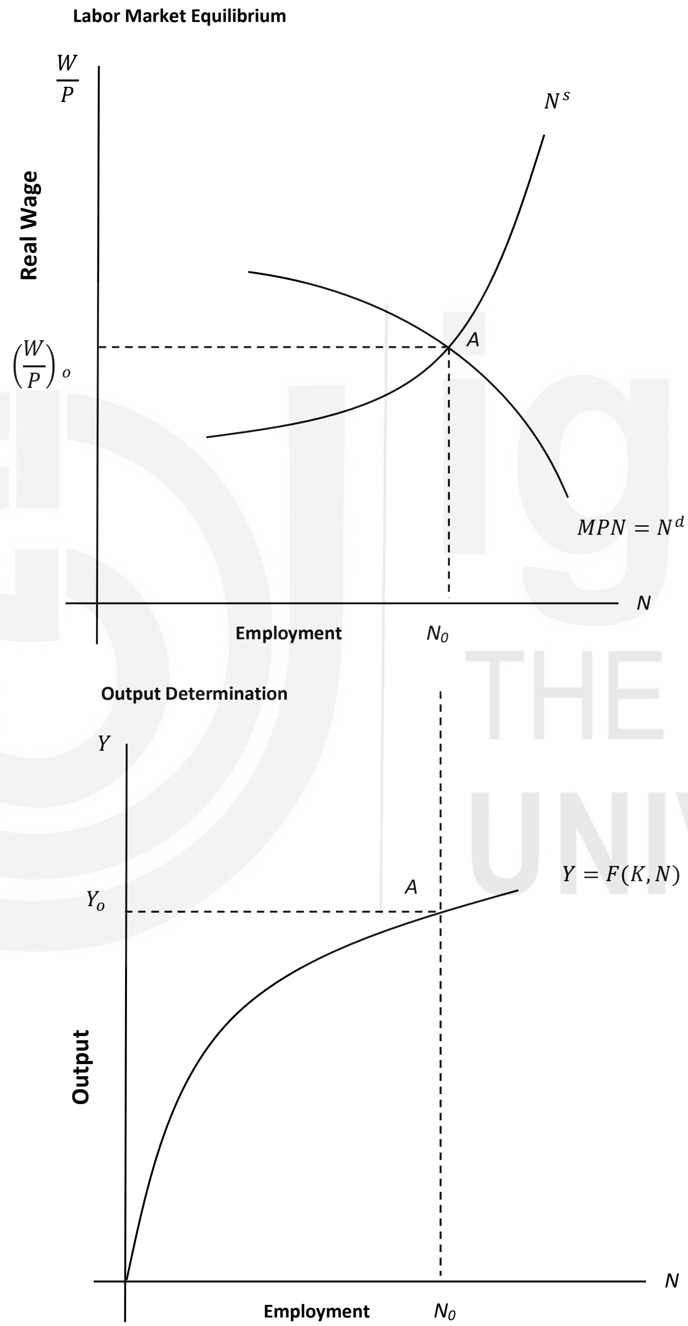

Fig. 1.2: Equilibrium Output and Employment

Since there is full employment in the economy all the time, with a given production function and capital, the Aggregate Supply (AS) is inelastic at full employment level of output. In Fig 1.3 we have shown the AS curve as a vertical straight line. Note that the AS curve will shift if there is a change in the level of technology (production function) or the level of capital.

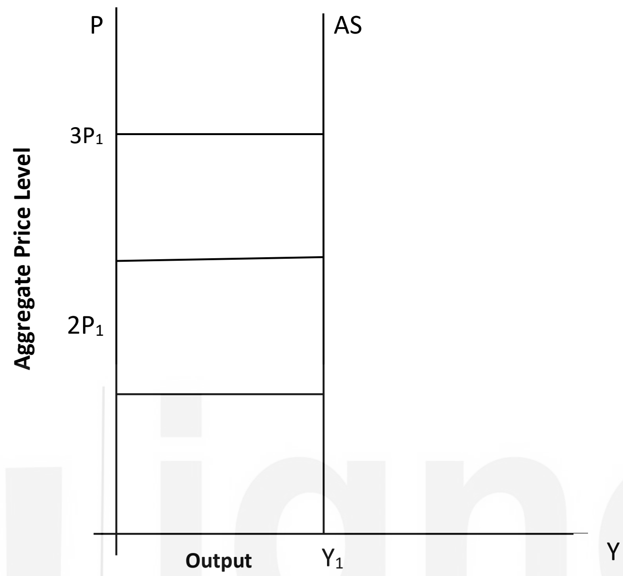

Fig. 1.3: Aggregate Supply Curve

1.5 QUANTITY THEORY OF MONEY

To understand the effects of changes in the stock of money, we need to study the money market equilibrium. Money is a stock variable. Its stock means its quantity at a point of time. Money is an asset demanded by the public for its holding. Money is supplied by the government and the banking system. The interaction of the demand for money and the supply of money create a money market. In this unit, we assume that the monetary authority autonomously determines the supply of money in an economy. Hence, the aggregate demand for money refers to the demand for money by the public. Thus, demand for money means the sum of all the money demanded by individual members of the public, whether households or firms.

There are various theories of demand for money, such as (i) the classical theory of demand for money, or the Quantity Theory of Money (QTM); (ii) the Keynesian theory of demand for money; and (iii) Friedman's restatement of Classical QTM. In this Unit, we will discuss the classical theory of demand for money. It is popularly known as the QTM, which is basically a theory of the price level. Recall that money has four functions; (i) a medium of exchange, (ii) a standard of deferred payment, (iii) a store of value, and (iv) a unit of account. According to classical economists, money is demanded for its function as a medium of exchange. There are several versions of the popular classical *QTM*. Below we present the transaction version, which is known as the Fisher's equation of exchange:

M.V = P.T … (1.5)

where,

T is number of transactions of average size, and proxy for income level

M is quantity of money supply,

V is velocity of circulation of money, and

P is the average price level.

Classical economists rely heavily on the QTM to establish the theory of demand for money. The QTM says that there is a proportional relationship between the amount of money held by the public and the price level. As per the classical assumption of full employment, the output level is given. Hence, T in equation (1.5) is fixed as a proxy for national income. Further, V is defined as the number of times a rupee moves from one hand to another during a given period. This means that it relies on the public's payment behaviour, which is constant in the long run.

The right-hand side of equation (1.5), PT represents the total amount of money required to facilitate transactions, i.e., the purchase or sale of total output in the economy. The left-hand side of equation (1.5) In the same equation, the left-side term MV is the product of the number of rupees in circulation and the number of times each is used for public payments. Hence, MV represents the total amount of money available for transactions during the given period. Equilibrium is achieved in the system at the point where money demanded (PT) equals money supply (MV).

The equation (1.5) can be rewritten as below,

$$P = \frac{V}{T} M \qquad \ldots (1.6)$$

As mentioned earlier, the terms V and T are constants in equation (1.6). Thus, the equation indicates that money supply directly affects price level. For instance, if M is increased by four times, price level P will rise by four times. Therefore, classical QTM is known as a theory of price level.

The other approach of classical QTM describes the relationship between the demand for money and 'nominal output' by taking directly real output instead of number of transactions. In equation form, it can be expressed as

MV = PY … (1.7)

where,

M is the money supply

V is the velocity of money that is the number of times the money changes hand.

P is the output price, and

Y is the real output level.

The total stock of money in the economy, measured by money supply times the velocity of circulation (MV), equals the nominal value of transactions or the nominal value of income or output in the economy (PY). The identity given at (1.7) is converted into the QTM under the assumption that **Y** and **V** are stable or constant in the short run. With **Y** and **V** being constant, the assumption that price level is passive means that **P** depends on changes in **M** rather than **M** depends on changes in **P**. These assumptions indicate that any short-run decrease (or increase) in **M** must lead to a proportional decrease (or increase) in **P**. Relaxation or invalidity of any one of these assumptions would not hold the proportionality relationship of P with M.

According to classical theory, under perfect competition with full employment in the economy, the output level is fixed ($\vec{Y}$). The velocity of circulation is predetermined ($\vec{V}$) and money supply is exogenously given. From equation (1.6), Price level (P) is proportional to the money supply (M).

$$P = (\vec{V}/\vec{Y}) \times M \qquad \ldots (1.8)$$

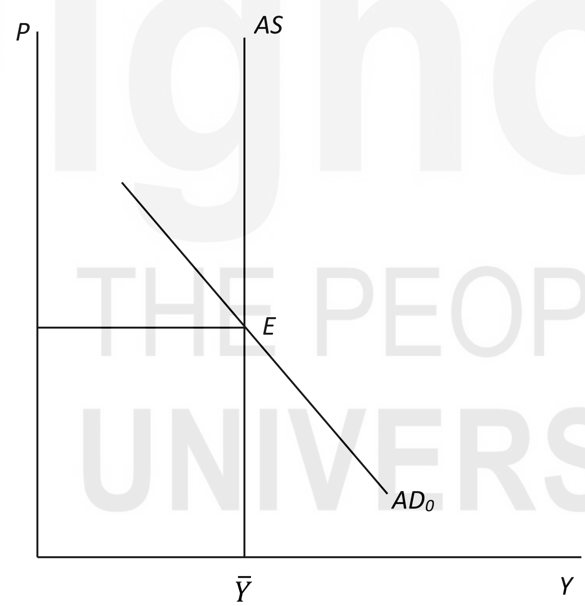

Fig. 1.4: Equilibrium Output and Prices

Thus, the labour market and production function determine the output level while money supply determines the price level in the classical scheme. The quantity theory of the money also gives the aggregate demand curve (see Fig. 1.4). Although the classical theory is accepted by most economists, Keynesian economists and Monetarists have criticized it. They believe that the classical theory fails in the short run when the prices are sticky. Also, it has been established that the velocity of money does not remain constant over

time. Nevertheless, the classical theory can be used to understand and control inflation in the economy.

1.6 SAY'S LAW OF MARKET

The Say's law of market has been a basic pillar of classical economic theory. According to this law, every supply generates its own demand. An implication of the above is that if a commodity is produced, it will create sufficient income for the owners of factors such as land, labor and capital to purchase the produced commodity. After all, the price of the commodity is divided into its components such as rent, wages and profits. Hence, this income will be sufficient to realize the price of the commodity.

According to the Say's law, enough income will always exist to purchase the entire produced output, so that there will not be any surplus left under the laissez-faire orientation. Using this logic, it can be concluded that there should not be any major depression. If it occurs, it must be caused by some interference with the free flow of commodities and money. The Say's Law thus supported the laissez-faire orientation of classical economics. The law is heavily dependent on the assumption of perfect flexibility in prices. Under this assumption, whatever quantity is produced can be sold in the market. In the short run, supply (or demand) may exceed demand (or supply). Such discrepancy is adjusted automatically, and instantaneously, through a decline (or rise) in prices. Such instantaneous adjustment exists for all commodities and factor inputs. Says' law is also regarded as a natural consequence of perfect competition.

The proponents of the Say's law believe that the law is valid both in a barter economy as well as a money economy. The law states that income received from the production of output is always spent on aggregate demand, i.e., consumption and investment. In other words, the law suggests that money is never hoarded and that the money or expenditure stream (MV) remains neutral. There is no doubt that the law is valid under a barter economy where production was primarily for consumption, i.e., whatever is produced is exchanged for goods and services. But one may doubt its validity in the present era, when production is based on future expectations and anticipations of demand; there is bound to be some overproduction. It is often said that the Say's law is valid if a substantial portion of production is used for consumption and the rest, if saved, is invested.

Check Your Progress 2

1) Describe how output and employment are determined in the classical model.

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2) What are the implications of the quantity theory of money?

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3) What is the relevance of the Say's law?

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1.4. CLASSICAL DICHOTOMY

From the quantity theory of money (QTM) we find the relationship between money supply and price level. We discuss below the effect of changes in money supply on other macroeconomic variables.

1.4.1 AD-AS Equilibrium

In an economy the real variables are real GDP, real capital stock and employment. These variables measure certain physical quantity. For instance, real GDP is defined as the quantity of goods and services produced in a specific period, such as a year, quarter or month. Similarly, real capital stock is defined as the quantity of machines and structures available at a specific time. The nominal variables are expressed in terms of monetary value, for example, the price level and the wage of a person in rupees.

Classical theory of output and employment suggests that the changes in the quantity of money affects nominal variables such as prices, nominal wages and nominal income. Changes in quantity of money, however, does not affect real variables. Classical economists provide an argument that real supply-side factors such as population, technology, capital stock, state of technology, marginal physical product of labour, and households' preferences regarding work and leisure determine real output and employment. The assumption of flexibility in prices leads to the self-adjustment of markets. Flexibility of prices and wages suggests that the changes in money supply affects the price level and nominal values such as money wages and nominal interest rates while real variables remain unaffected. Such independence of real economic variables from changes in money supply and nominal variables is known as 'classical dichotomy'.

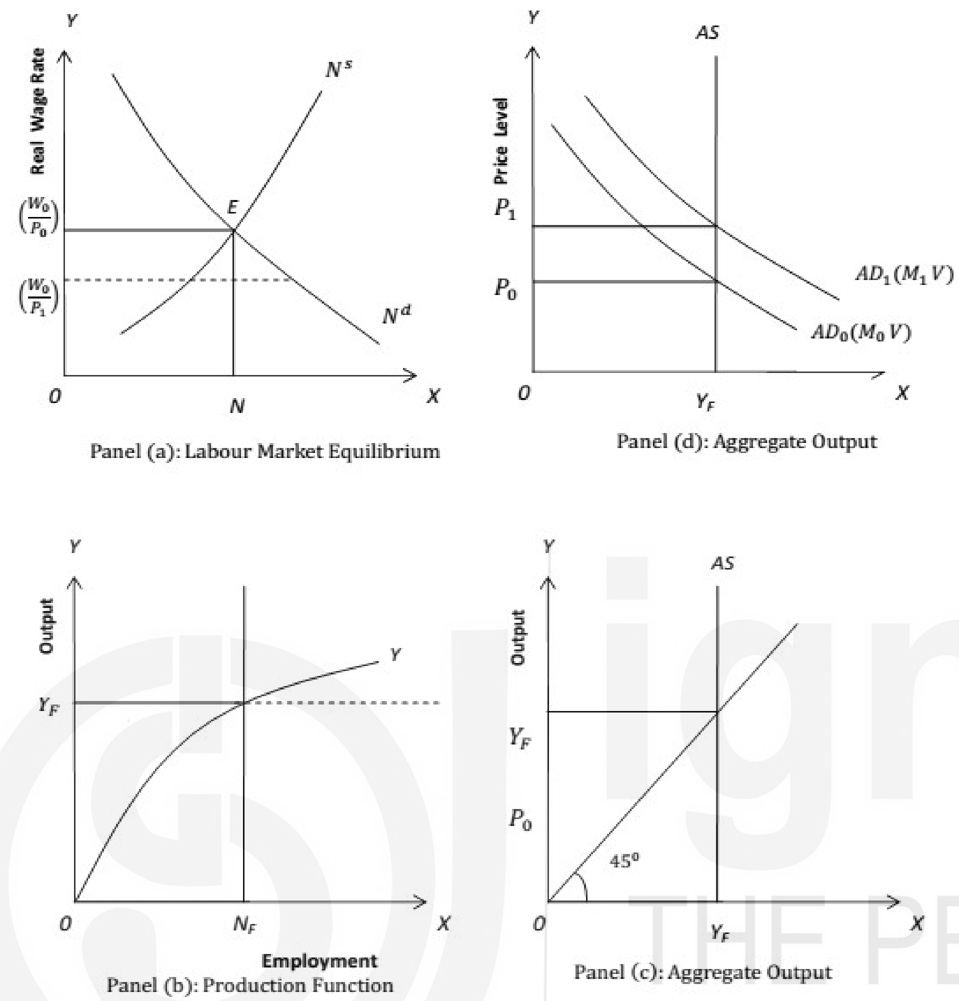

Fig. 1.5: Classical Model of Output Determination

1.4.2 Neutrality of Money

An implication of classical dichotomy is that 'money is neutral'. We have illustrated 'neutrality of money' graphically in Fig. 1.5. There are four panels in Fig. 1.5. Suppose money supply in an economy is equal to M_0 at a specific time. The corresponding aggregate demand curve is AD_0 as seen in panel (d) of Fig. 1.5. Its interaction with the aggregate supply curve AS decides the price level P_0. Corresponding to the price level P_0, the labour market determines the equilibrium money wage rate W_0. Hence, the real wage rate is equal to W_0 / P_0 and equilibrium level of employment is N_F as seen in Panel (a) of Fig. 1.5. Given the production function (see Panel (b)), with the equilibrium level of employment N_F the aggregate output is Y_F. Let us assume that the monetary authority of the country expands its money supply from M_0 to M_1 which results in an upward shift in the aggregate demand curve to AD_1 as seen in Panel (d) of Fig. 1.5. As a result, the price level increases from P_0 to P_1. This causes the real wage to fall from (W_0/P_0) to (W_0/P_1) as seen in Panel (a) of Fig. 1.5. At the real wage decreases to (W_0/P_1), the demand for labour increases. Thus, labour demand exceeds labour supply. According to classical theory, this causes the nominal wage rate to increase to W_1 (in equal proportion to the increase in price level) such that the original level of real wage is maintained (that is, $W_1/P_1 = W_0/P_0$). Thus, there is no change in the equilibrium level of employment N_F. (See panel (a) of Fig. 1.5). At the earlier wage rate and employment level, the real output is not affected, as seen in panel (b) of Fig. 1.5. To conclude, when there is an expansion of money supply, the nominal wage rate and price level increase without affecting the levels of real wage, employment and output. Such independence of real economic variables from changes in money supply is called neutrality of money.

There is a crucial limitation to the neutrality of money. It is a basic result derived from the full-employment equilibrium that is based on the full flexibility of prices. If increase in money supply (resulting in higher prices) had no real effects, then inflation would not have been a serious concern in an economy. Inflation, however, is a serious problem as has many adverse effects such as decrease in quality of life of people, decrease in economic growth, etc. Thus, measures are taken to control inflation and maintain price stability in an economy.

1.4.3 Saving-Investment Equilibrium

You should note that classical theory emphasized on the function of money that it is a medium of exchange. According to the classical school money is demanded for transaction purposes only. Hence, supply of and demand for money in the classical theory do not determine the equilibrium rate of interest. An increase in the quantity of money does not affect the real rate of interest. Thus, there is no change in the level of saving and investment (see, Fig. 1.6). This shows that when money supply rise, it does not disturb the capital market equilibrium (or saving-investment equality) and the level of full-employment equilibrium remains unchanged.

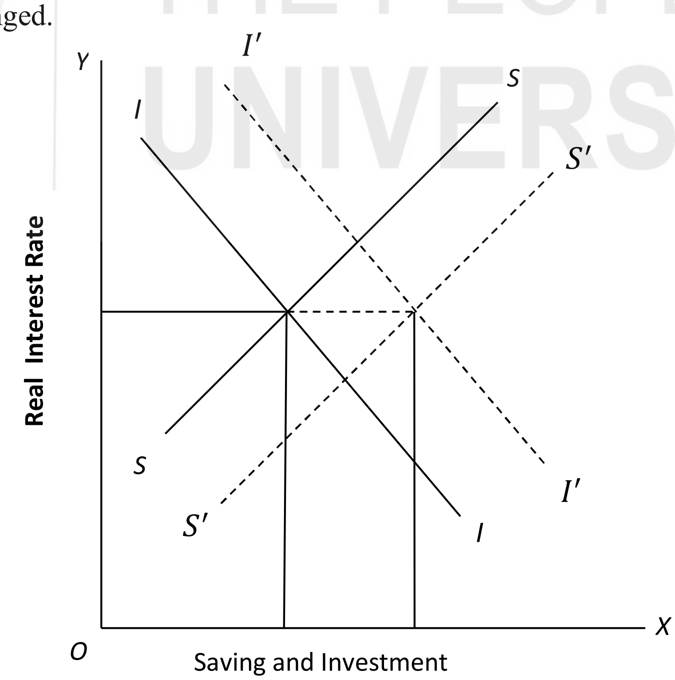

Fig. 1.6: Capital Market Equilibrium

An increase in money supply will lead to an increase in prices. Due to the higher prices, nominal investment expenditure will increase. However, according to classical theory, such increase will be proportional to the increase in prices. Therefore, investment expenditure in real terms will not change. This is explained diagrammatically in Fig. 1.6. We measure real interest rate on the y-axis and nominal value of saving-invest on the x-axis. The increase in money supply causes the supply curve of nominal saving to shift downward to the right from SS to $S'S'$. Simultaneously, the investment demand curve will shift upward from II to $I'I'$ by the same amount. It implies that the interaction of $S'S'$ and $I'I'$ determines the interest rate without affecting real saving and real investment at the higher price level too.

1.5 LONG RUN VS. SHORT RUN

In general, supply and demand fluctuate for various reasons, which affects the level of output. There is considerable difference between short-run and long-run fluctuations in output. Most economists believe that the behavior of processes is different between the short run and the long run. In the long run, prices are flexible, which may be affected by changes in supply or demand. In the short run, most of the prices are "sticky'' at the pre-determined level. As the behaviour of prices in the short run is different from that in the long run, the effects of economic events and policies vary over different time horizons. Classical theory is considered to be applicable in the long run. In the short run, we may need a different analytical framework. In Unit 2 we discuss the Keynesian model, which is more applicable in the short run.

Let us explain the price behavior in both the short-run and long-run in Fig. 1.7. Aggregate supply is the total amount of goods and services that firms sell in an economy. Aggregate demand is the total quantity of goods and services that are purchased. In a standard AS-AD model, output (Y) is presented on the X-axis and price (P) is on the Y-axis. The long run aggregate supply curve (AS_{LR}), which denotes the potential output (full employment output) of the economy is given by a vertical line at Y^*. The short-run supply curve is upward sloping (AS_0). The interaction of supply and demand determines the equilibrium level of output (Y_0). In the short run, an outward shift in the supply curve (from AS_0 to AS_1) results in rising output (from Y_0 to Y_1) and falling prices (from P_0 to P_1). An outward shift in the AD curve (from AD_0 to AD_1) also results in rising output (from Y_0 to Y_2) but rising prices (from P_0 to P_2).

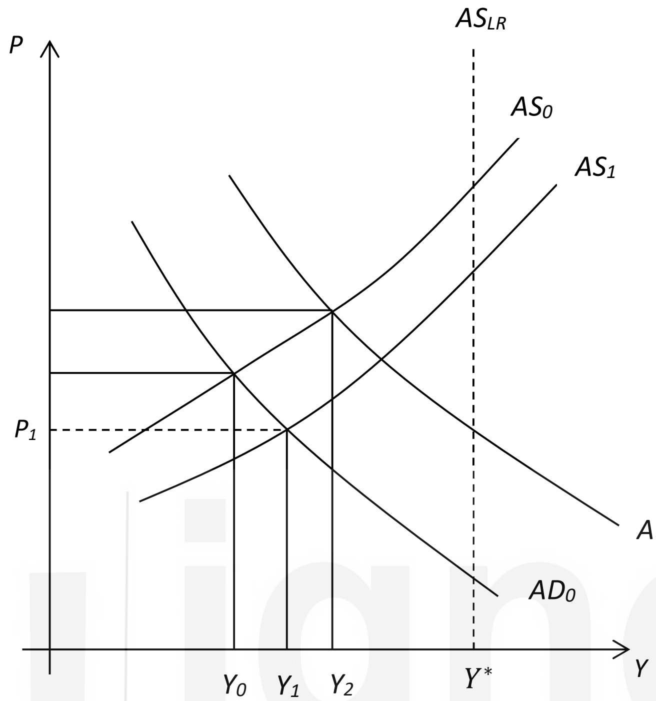

Fig. 1.7: Determination of Prices in the Long-Run and Short-Run

Let us look into certain other situations. Short-run fluctuations in nominal variables usually result in a change in the output level, as macroeconomic variables not fully flexible. An increase in money supply, for example, will shift the AD curve to the right (i.e., aggregate demand will increase). This will increase prices and income. In contrast, a decrease in money supply will result in a shift in the AD curve to the left, which will decrease both prices and income. According to the classical theory an increase in money supply leads to increased prices without affecting the real output. This may be applicable in the long run, not in the short run.

Check Your Progress 3

1) What is meant by classical dichotomy?

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2) Why is money considered to be neutral? What are its implications?

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3) Is classical theory applicable in the short run?

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1.6 LET US SUM UP

In this unit, we discussed different schools of macroeconomic thought that have evolved over time to respond to certain events in the world economy. Relying on the assumption of full flexibility of wages and prices, classical theory concludes that the economy always tends to equilibrium at the full employment level of output. Also, the capital market always ensures that there is enough demand in the market. The classical quantity theory of money determines the level of prices and wage rates without affecting the real economic variables. Classical theory advocates that policy intervention cannot influence output levels in an economy as it is self-adjusting. In the classical framework money considered to be neutral in the sense that it does not affect real variables.

1.7 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) In Section 1.3 we have given the important features of classical theory. Go through it and answer.

2) New-classical economics considers a general equilibrium framework, microfoundations and rational expectations.

Check Your Progress 2

1) Go through Sub-Section 1.4.1. Refer to Fig. 1.5 and describe the process.

2) The quantity theory of money (MV = PY) establishes the relationship between money supply, velocity of money, price level and output. When the velocity of money and the output level remain constant, an increase in money supply will lead to increase in prices.

3) According to the Say's law, supply creates its own demand. It rules out the possibility of unemployment in an economy.

Check Your Progress 3

1) It refers to the classical proposition nominal variables do not influence real variables in an economy. It arises from the classical assumptions of perfect competition and full flexibility in prices and wages.

2) Classical economists emphasized that money is demanded as a medium of exchange. An increase in money supply leads to proportionate increase in prices and wages; real variables such as output and employment remain unchanged.

3) Prices and wages may not be fully flexible. Rigidities in prices and wages may not lead to instantaneous adjustment in output and employment. Thus, classical theory may not be applicable in the short run.

UNIT 2 THE KEYNESIAN MODEL*

Structure

20.0 OBJECTIVES

After going through this Unit you should be in a position to

- appreciate the assumptions underlying the Keynesian model;
- explain the concepts of consumption function and saving function;
- identify the factors affecting investment demand;
- show that equilibrium output in the Keynesian model is determined by the level of aggregate demand;
- explain the various types multipliers in the Keynesian system;
- explain the paradox of thrift;
- explain the concepts of liquidity preference and liquidity trap; and
- discuss the Keynesian view on the role of the government in the economy .

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* Prof. Ananya Ghosh Dastidar

2.1 INTRODUCTION

In this Unit we will learn about the Keynesian model that was first developed by the British economist John Maynard Keynes in his famous book 'The General Theory of Employment, Interest and Money', published in 1936. This model differs in several respects from the classical approach to macroeconomics that you studied in the previous Unit. The Keynesian model allows for the existence of unemployment as well as rigidity in wages and prices. It recognizes that lack of effective demand may prevent the attainment of full employment of resources. Also, it emphasizes the need for government intervention – as the market mechanism would not be able to restore full employment equilibrium due to rigidities in wage rate and prices.

2.2 COMPONENTS OF AGGREGATE DEMAND

We observe that Keynesian economics gained importance against the backdrop of the Great Depression of the 1930s, when developed economies like the USA and Great Britain faced a deep and prolonged recession, with rising unemployment, falling revenues and profits of firms and rise in unutilized capacity. In this situation the market mechanism failed to restore equilibrium and the Keynesian policy of raising aggregate demand via increase in public expenditure played a crucial role in addressing the problem.

Aggregate demand is the planned expenditure on goods and services by all the economic agents, such as households, firms and government. There are three main components of aggregate demand in a closed economy: (i) *planned* consumption expenditure (C) of households, (ii) *planned* investment expenditure (I) of firms, and (iii) expenditure incurred by the government on goods and services (G). Throughout our discussion in this Unit, each of C, I and G are assumed to be in real terms (i.e., no change in prices).

2.2.1 Consumption Function

Keynes proposed the idea of a Consumption function which gives the relation between *planned* consumption expenditure of households (C) and aggregate income (Y). It can be represented by the following linear equation:

$$C = a + bY \quad \ldots (2.1a)$$

In the presence of the government, a part of income would be paid in taxes (T). In that case the consumption function would be represented as a function of disposable incomes ($Y_d = Y - T$), that is,

$$C = a + b(Y - T) = a + bY_d \quad \ldots (2.1b)$$

For the discussion below, for simplicity, we assume that there is no government, so that T = 0 and Y = Y_d.

The slope of the consumption function 'b' is assumed to be a positive fraction (i.e., 0 < b < 1). It means that the consumption function can be represented as an upward sloping straight line as in Fig. 2.1. It indicates that planned consumption expenditure increases with an increase in the level of aggregate income.

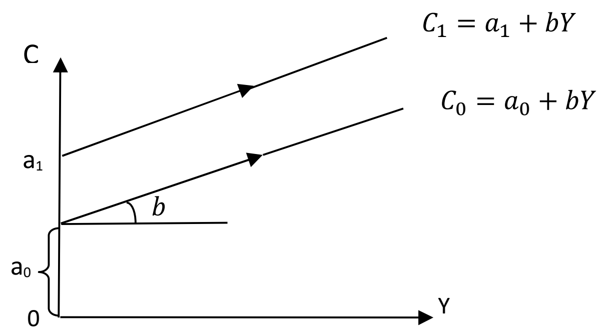

Fig. 2.1: Consumption Function

Note that 'b' is defined as the *marginal propensity to consume* (MPC) and it represents the change in consumption induced by a unit change in income. The MPC is assumed to be less than one. If there is an increment in income by ΔY, the induced increase in consumption ΔC is a fraction of ΔY (i.e., $\Delta C = b\Delta Y$ and $\Delta C/\Delta Y = b < 1$). It means that the entire increment in income is **not** spent on consumption, a fraction $(1 - b)$ is saved. It means total saving out of the incremental income is $(1 - b)\Delta Y$.

Let us consider the case where the consumption function is given by C = 20 + 0.25Y. If aggregate income increases by Rs.100 (i.e., ΔY=100), then Rs. 25 (ΔC = 0.25×100) is the additional consumption and Rs.75 [(1 – 0.25) ×100] is additional saving of households out of the incremental income. Similarly, if income declines by Rs.100, then consumption would decline by Rs. 25 only, while saving would decline by Rs. 75.

The intercept of the consumption function 'a' represents autonomous consumption, with $a > 0$. From equation (2.1a), you can see that when $Y = 0$, we have $C = a$. It implies that consumption occurs even when income is zero. This consumption is financed by drawing down past saving or by 'dis-saving'. An increase in 'a' represents increase in planned consumption of households at each level of income; this can happen, for instance, when consumer confidence increases (i.e., households' expectations about the future turn buoyant). You can see in Fig. 2.1 that an increase in the intercept (from a_0 to a_1) results in a parallel, upward shift of the consumption function, from C_0 to C_1.

Aggregate consumption has two components: (i) an autonomous component 'a' that does not depend on income, and (ii) an 'induced' component 'bY' that depends on aggregate income levels. An increase (decrease) in Y induces an increase (decrease) in planned consumption expenditure by households.

The ratio of aggregate consumption expenditure to aggregate income $\left(\frac{C}{Y}\right)$ is defined as the *average propensity to consume* (APC). From equation (2.1a) you can see that APC = $C/Y = a/Y + b$. Note that APC will decline as Y increases, while MPC remains constant.

2.2.2 Investment Function

The investment function (I) represents *planned* investment expenditure undertaken by firms on machinery, factories and other such items that add to productive capacity of the economy. In the simplest version of the Keynesian model investment is assumed to be 'autonomous', i.e., independent of the level of aggregate income (Y). In this case the investment function is represented by the equation as below.

$$I = \bar{I} \qquad \ldots (2.2a)$$

In equation (2.2a) a 'bar' over the variable investment ($\bar{I}$) variable indicates a fixed amount. However, we can easily relax this assumption and allow investment to be a function of aggregate output. In this case the investment function is:

$$I = c + d\,Y \qquad \ldots (2.2b)$$

where c > 0 and *d* > 0; 'c' is the autonomous part of investment, *d* is the *marginal propensity to invest* (MPI) and '*d* Y' represents investment that is induced by aggregate income.

Fig. 2a below depicts an autonomous investment function, while Fig. 2.2b depicts an induced investment function which has a component that is induced by income.

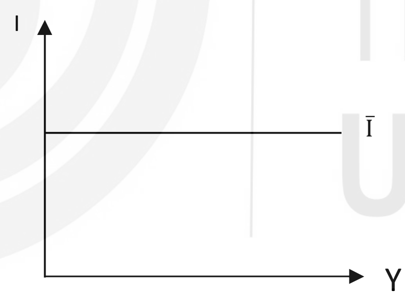

Fig. 2.2a: Autonomous Investment

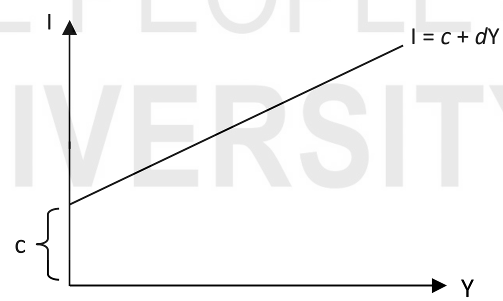

Fig. 2.2b: Induced Investment

Note that the term investment refers to planned expenditure by firms on new production equipment (e.g., machinery, building, etc.) that add to productive capacity. It does *not* refer to buying of corporate bonds or shares in a company or opening a fixed deposit account in a bank.

An important variable that affects firms' decision to invest is the rate of interest. As the rate of interest increases, the cost of borrowing goes up, resulting in a reduction in the firms' planned investment expenditure. The inverse relation between investment and interest rate can be further explained as follows:

Say, the current interest rate is i and an investment project that costs Rs. A, is expected to last for T periods. In each period it yields a return of R_t, where t = 1,2,…, T. In this case, the present value of the stream of returns from the project P is calculated as:

$$P = \frac{R_1}{(1+i)} + \frac{R_2}{(1+i)^2} + \frac{R_3}{+(1+i)^3} + \cdots + \frac{R_T}{(1+i)^T} \qquad \text{… (2.2c)}$$

If profit is the only motive of investment, this project will be taken up only if $(P - A) \geq 0$.

For example, if you are willing to lend Rs.100 at 10 per cent rate of interest for a period of 1 year, then at the end of the year you will receive Rs.110. So, at the current interest rate of 10 per cent, the *present discounted value* of Rs.110 is Rs. 100 [where, 100 = 110 / (1+0.1)].

Now, for the economy as a whole, for any given rate of interest, i_0, the corresponding level of investment I_0 by firms would consist of all the projects for which $P \geq A$. Now, if the interest rate *increases* to i_1 (R_t and A remaining unchanged), P would *decrease.* Consequently, some projects for which $P \geq A$, would now be unprofitable. At a higher interest rate, therefore, all the projects for which P < A would not be taken up, so there would be a fall in the associated level of investment I_1. Thus, when $i_0 > i_1$ we have $I_0 < I_1$.

If we include interest rate as an explanatory variable in the investment function, we have

$$I = c + dY - e\, i \qquad \text{… (2.2d)}$$

where c > 0, d > 0 and e > 0. Here, e captures the interest responsiveness of planned investment expenditure.

Check Your Progress 1

1) Discuss the main features of the Keynesian consumption function.

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2) Through a diagram, explain how (i) a decrease in autonomous consumption, and (ii) an increase in the MPC, would affect the consumption function.

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3) Distinguish between autonomous investment and induced investment.

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4) State the reasons for the inverse relationship between interest rate and investment level.

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2.3 DETERMINATION OF OUTPUT IN THE KEYNESIAN MODEL

The Keynesian model is based on certain assumptions. First, there are unemployed resources in the economy. In other words, there is underutilized productive capacity (e.g., factories operating below capacity, unused machines, labour without employment, etc.). It follows from this that output in such an economy is demand determined, i.e., existing production capacity can be used to step up production in case there is an increase in aggregate demand for output. Second, money wages and prices are not flexible (i.e., there are price rigidity and wage rigidity). Since there is unemployment and workers are willing to work at the going wage rate, an increase in the supply of output is possible at the given price level. Third, output is assumed to be a homogenous commodity used both for consumption and investment.

We will present the simplest Keynesian model applicable to a three-sector economy consisting of households, firms and the government. The economy is closed (i.e., no external trade). Later on we relax this assumption and allow for external trade.

In the three-sector economy, aggregate demand comprises (i) planned consumption by households (C), (ii) planned investment by firms (I), and (iii) expenditure on goods and services by the government (G). Initially we assume that investment is fully autonomous, an assumption that is relaxed later. The government is assumed to raise taxes and here we consider the cases of both lump sum taxes as well as proportional taxes. Later we will also consider the implications that arise when the government makes transfer payments to households (e.g., pensions, unemployment dole, etc.).

Therefore, in our simple Keynesian model

Aggregate Demand (AD) = C + I + G

By substituting the values of C, I and G, we obtain

$$AD = a + b(Y - T) + \bar{I} + G \qquad \dots (2.3)$$

Here T is a lump sum tax levied by the government; the investment function is given by (2.2a).

2.3.1 Determination of Equilibrium Output

The equilibrium condition in the Keynesian model is that aggregate demand for output is equal to aggregate supply (Y) or, the total output produced in the economy. Note that at equilibrium, the plans of households and firms (regarding expenditure on consumption and investment) are also fulfilled. In case plans are *not* fulfilled there would be changes in aggregate demand (as households and firms would adjust their plans) and hence output produced would change as well, so that the situation would *not* be one of equilibrium.

The equilibrium condition in the economy is achieved when

Aggregate Supply = Aggregate Demand

From equation (2.3) we know that

$$Y = a + b(Y - T) + \bar{I} + G$$

Solving this for Y, equilibrium output (Y_e) can be obtained as:

$$Y = a + bY - bT + \bar{I} + G$$

Or,

$$Y - bY = a - bT + \bar{I} + G$$

Thus,

$$Y_e = (a - bT + \bar{I} + G)/(1 - b) \qquad \dots (2.4)$$

From (2.4) it should be clear to you that equilibrium output would be higher at higher levels of (i) autonomous investment, (ii) government expenditure, and (ii) the MPC. Recall that 0 < MPC < 1, so that the term (1– b) in the denominator is always positive.

We will now carry out graphical analysis to further understand the process of attainment of equilibrium in the simple Keynesian model. First we will consider the case of a two-sector economy with only households and firms and no government. Later we will consider a three-sector economy which has the government sector as well.

Two-Sector Economy without Government

In a two-sector economy, there are only households and firms. Therefore, aggregate demand has only two components, C and I. Thus, $AD = C + \bar{I}$. Setting G = T = 0 in equation (2.4) above, we see that equilibrium output would be

$$Y_e = (a + \bar{I})/(1 - b) \qquad \ldots (2.5)$$

We depict equilibrium level of output in Fig. 2.3. Along the horizontal axis we measure output or income (recall that income is generated in the process of production of output, so we use the terms income and output interchangeably). Along the vertical axis we measure aggregate demand and its components (C and I).

Note that OL is a line which makes an angle of 45^0 with the horizontal axis, so that a perpendicular from this line such as AY_1, forms an isosceles triangle AOY_1, with $\angle OAY_1 = \angle AOY_1 = 45^0$, and as per the property of isosceles triangles, the two sides $O\,Y_1$ and $A\,Y_1$ are equal. Using this construct we are able to compare vertical distances (demand) with the horizontal distances (output or income). Note that at A, where the consumption function cuts OL, consumption expenditure AY_1 is exactly equal to income OY_1, so that saving is zero.

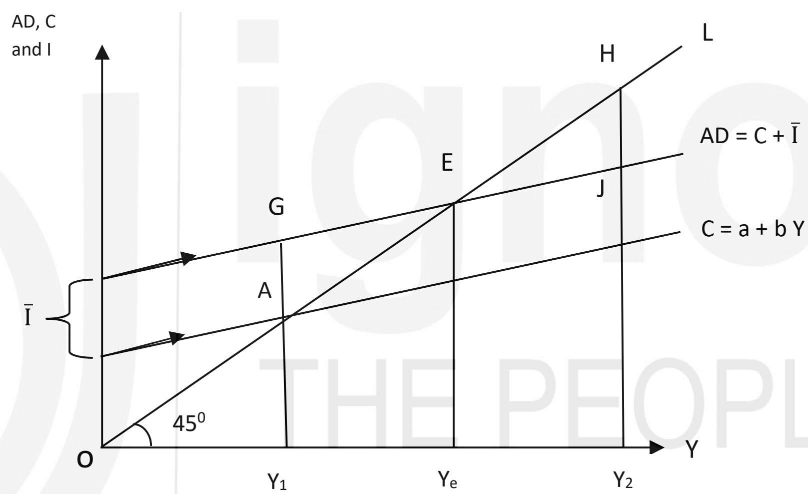

Fig. 2.3: Equilibrium Output in a Two-Sector Economy

In Fig. 2.3 we derive the aggregate demand curve (AD) by adding the investment ($\bar{I}$) and consumption (C) functions. Since investment is autonomous the line AD is parallel to the consumption function, i.e., slope of AD is the same as the slope of the consumption function. Equilibrium output is attained at E, at the intersection of AD with OL where aggregate demand EY_e is equal to aggregate output OY_e. Note that the equilibrium is unique, as there is only one level of income OY_e at which aggregate demand is equal to aggregate supply. In this case firms will be able to sell exactly as much as they had planned and hence there will be no unplanned changes in their stock of inventories.

For all income levels less than OY_e, i.e., to the left of E, there is excess demand for output. For example, when aggregate income is OY_1 ($=AY_1$), the corresponding level of aggregate demand GY_1 is greater than the supply of output AY_1 ($=OY_1$). In this case since demand is more than the output produced by firms, they will have to run down their stock of inventories. So in the next period, firms will increase production and output would move toward OY_e. This process of increase in output (and hence income) will continue till OY_e is reached, where the amount produced (OY_e) being exactly equal to the amount demanded (EY_e), there is no unplanned changes in the stock of inventories and hence no further change in production levels.

Similarly, to the right of E (i.e., at all income levels greater than OY_e), there will be excess supply of output. When income is OY_2, the corresponding level of aggregate demand JY_2 is less than the supply HY_2 (= OY_2). In this case, firms will be unable to sell the entire amount produced ($HY_2 > JY_2$) and hence their stock of inventories will increase (i.e., there will be unplanned inventory accumulation by firms). This would induce firms to reduce production in the next period. This process will continue till the equilibrium output EY_e (=OY_e) is reached, where demand and supply are equal and there are no more unanticipated changes in firms' stock of inventories.

From this discussion we see that output market equilibrium in the Keynesian model is stable. Any situation of disequilibrium tends to be self-correcting, inducing a movement back towards the equilibrium level of output.

Three-Sector Economy with Government

In equation (2.4) we derived equilibrium output in a three sector model, with households, firms and a government sector. Diagrammatically this can be shown in Fig. 2.4. As earlier, the OL line makes an angle of 45^0 with the horizontal axis along which we measure income (or output). Aggregate demand is measured along the vertical axis. Now, this has three components, C, I and G, where investment and government expenditure are autonomous.

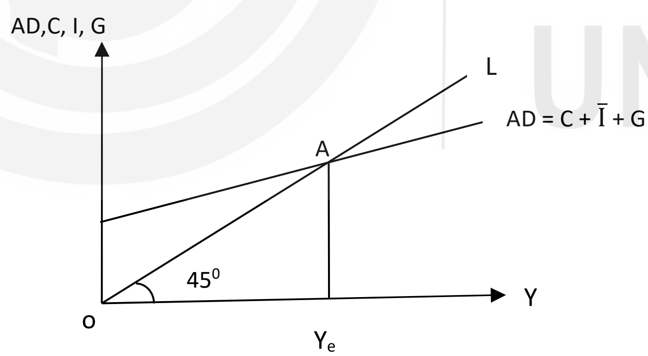

Fig. 2.4: Equilibrium Output in a Three-sector Economy

Equilibrium in the three-sector model occurs at the intersection of the AD curve and the OL line, where demand AY_e is equal to aggregate supply OY_e. You should see that equilibrium income level OY_e is unique and that this equilibrium is stable. At all income levels higher than OY_e there is excess supply, so that firms decrease production, inducing a change in output towards OY_e. On the other hand, at all income levels lower than OY_e, there is excess demand, inducing firms to increase production.

Note that the position of the AD curve is determined by the size of the autonomous components of aggregate demand (viz., I, G and autonomous consumption *a*). The higher the levels of these components, the higher would be the AD line and the higher would be the equilibrium output level (Y_e), implying that output is demand-determined in the Keynesian model.

2.3.2 Investment Multiplier

We will now examine the comparative static properties of the Keynesian model and explain the concept of the multiplier.

Autonomous Investment

Consider once again the two-sector model with only firms and households, such that equilibrium income is $Y_e = (a + \bar{I})/(1 - b)$. If, *ceteris paribus,* there is a change in the level of autonomous investment (ΔI), the associated change in income would be given by $\Delta Y = (1/(1 - b) \times \Delta I$. Thus, $\frac{\Delta Y}{\Delta I} = (1/(1 - b)$. Here, we consider change in one component of autonomous demand at a time. This means when $\Delta I > 0$, we shall assume that $\Delta a = 0$ and so on. The term $(\frac{1}{1-b})$ is greater than one, since our assumption is that $(0 < b < 1)$. An implication of the above is that when investment increases by ΔI, the associated increase in income is greater than ΔI, as it is a multiple of ΔI. How is this possible? To see this, consider the several rounds of increments in expenditure that are triggered by a onetime increase in autonomous expenditure.

If we increase I by ΔI, income will increase by ΔI in the first round, which will induce a second round increase in consumption by $b\Delta I$ (since consumption is a function of income). Since output is demand-determined, this increase in consumption demand raises output and hence income increases further by $b\Delta I$. This induces a third round increase in consumption by $[b(b\Delta I) = b^2\Delta I]$ which triggers further increases in income ($b^2\Delta I$). In the fourth round, consumption demand increases by $[b(b^2\Delta I) = b^3\Delta I]$. The process continues with increase in consumption raising output, income and yet another round of consumption. Note that since MPC < 1 (i.e., $b < 1$), each subsequent round of increase in consumption is weaker than the previous one, so that the entire process dies down gradually.

Therefore, the total increase in income due to a onetime increase in autonomous investment is:

$$\Delta Y = (\Delta I + b\Delta I + b^2\Delta I + b^3\Delta I + \ldots..) = \Delta I\ (1 + b^2 + b^3 + b^4 + \ldots..) \quad \ldots(2.6)$$

Since $0 < b < 1$, the expression in parentheses on the right hand side of (2.6) can be expressed as the sum of a convergent infinite geometric progression series, i.e., $1 + b^2 + b^3 + b^4 + \ldots.. = 1/(1 - b)$. Therefore, $\Delta Y = [1 / (1 - b)]\ \Delta I$, which shows that income increases by a multiple of the initial increase in autonomous investment and the multiplier is $(\frac{1}{1-b})$. This is shown diagrammatically in Fig. 2.5.

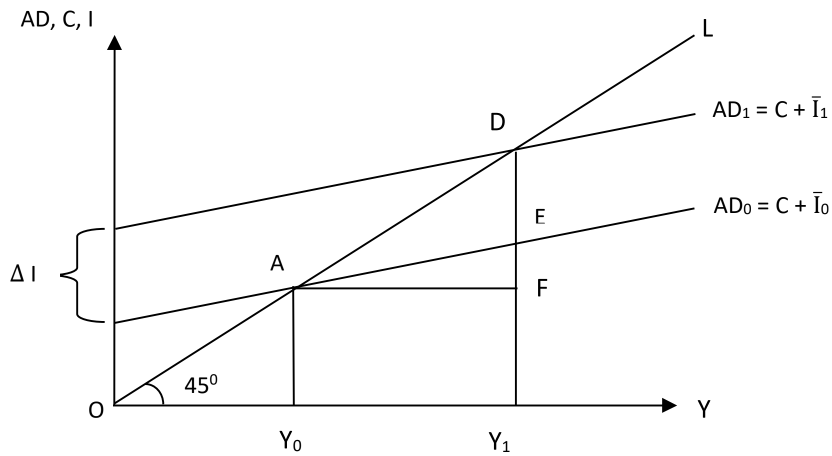

Fig. 2.5: Investment Multiplier

The initial equilibrium in Fig. 2.5 occurs at A where AD_0 (= C + $\bar{I}_0$) intersects OL. With an increase in autonomous investment to $\bar{I}_1$ (i.e., $\Delta I = \bar{I}_1 - I_0$), the AD curve shifts upward to AD_1 (= $C + \bar{I}_1$) and the new equilibrium occurs at D where AD_1 intersects OL. Note that while investment increased by ΔI (= DE), the change in income ΔY (=DF) that it brings about is greater, i.e., $\Delta Y > \Delta I$ (where, $\Delta Y = Y_0Y_1$= AF = DF and DF > DE = ΔI), indicating the operation of a multiplier effect.

Induced Investment

Suppose the investment function is given by equation (2.2b) above, i.e., $I = c + dY$. In this case, investment demand has an induced component that is affected by changes in income. The equilibrium income would be derived as follows:

$$Y = AD = a + bY + c + dY, \text{ so that:}$$

$$Y_e = (a + c) / [1 - (b+d)] \qquad \dots (2.7)$$

ceteris paribus, if there is a change in autonomous investment in this case (i.e., $\Delta I = \Delta c$; $\Delta a = 0$), the multiplier would be given by: $1 / [1 - (b+d)]$. This would follow since $\Delta Y = \{1 / [1 - (b+d)]\}\Delta c$. Note that in this case we assumed that $(b + d) < 1$ (i.e., the sum of MPC and MPI is less than one).

2.3.3 Government Expenditure and Tax Multipliers

We now consider the case of a three-sector economy. For simplicity we assume that investment demand is autonomous and the government incurs a constant amount of expenditure G which is autonomous. The government also collects taxes T. We will consider two cases: (i) the government collects a lump sum tax T, and (ii) the government sets a proportional tax rate t, so that its tax revenue $T = tY$. The government's budget deficit (B) is defined as $B = G - T - TR$, where TR refers to transfer payments made by the government. In what follows we will assume, for simplicity that $TR = 0$.

Lump Sum Taxes

When the government imposes a lump sum tax *T*, the equilibrium income would be given by (2.4) above: $Y_e = (a - bT + \bar{I} + G) / (1-b)$. In this case, the multiplier for any change in autonomous expenditure (either investment or government expenditure) would be given by: $1 / (1 - b)$. Thus, $\Delta Y/\Delta G = 1 / (1 - b)$ or $\Delta Y/\Delta I = 1 / (1 - b)$. If there is an increase in the amount of tax collected, then income would fall. In such a case the tax multiplier would be given by: $-b / (1 - b)$. Thus, $\Delta Y/\Delta T = - b / (1 - b)$. Recall that taxes are a leakage from the income stream, so increase in taxes leads to a decrease in disposable income. Consequently, there is a decrease in consumption demand, leading to a decline in output (as output is demand-determined, lower demand results in lowering of output). You should note that there is a two-way relationship: consumption is influenced by the level of income (evident from the consumption function) and income is influenced by consumption demand.

Balanced Budget Multiplier

Suppose the government increases its spending (i.e., $\Delta G > 0$) and cut taxes by exactly the same amount ($\Delta T < 0$). It implies that the government budget remains unchanged. If ΔB indicates budget balance, then $\Delta B = \Delta G - \Delta T = 0$. Alternatively we can write $\Delta G = \Delta T$. You may think that such action would have no impact on equilibrium income as the injection of additional demand (ΔG) is exactly offset by a leakage from the demand stream ($\Delta G = \Delta T$). However, we can show that the multiplier in this case is equal to one, i.e. income increases by the amount of increase in government expenditure. From equation (2.4) we know that $Y_e = (a - bT + \bar{I} + G) / (1-b)$, so that $\Delta Y = (- b\Delta T + \Delta G) / (1- b)$. Since $\Delta G = \Delta T$, this can be written as $\Delta Y = (1 -b) \Delta G / (1- b)$, or $\Delta Y / \Delta G = 1$.

Proportional Taxes

Now suppose the government levies a proportional tax on income at tax rate *t*, where $0 < t < 1$. Since $AD = a + b (Y - tY) + \bar{I} + G$, the equilibrium condition is given by

$$Y = a + b (1 - t)Y + + \bar{I} + G \qquad \ldots (2.7)$$

By re-arranging terms, we obtain the equilibrium level of output as

$$Y_e = (a + \bar{I} + G) / \{1 - b (1 - t)\} \qquad \ldots (2.8)$$

Equation (2.8) implies that the autonomous expenditure multiplier (for a change in investment or government expenditure) is now lower compared to the cases without the proportional tax. Without proportional taxes the multiplier is given by $1/(1 - b)$. Now, in the presence of proportional taxes, it is given by $\Delta Y / \Delta G = \Delta Y / \Delta I = 1 /\{1 - b (1 - t))$. The reduction in the value of the multiplier arises because, out of the increase in income in each round, a fraction *t* has to be paid in taxes.

You can see from equation (2.8) that any increase in the tax rate would lower equilibrium income, whereas any decrease in tax rate would have the opposite effect. Therefore, you should see the logic behind tax cuts offered by governments to revive economic activity. However, remember that our assumption is that there is excess production capacity, which is the main reason why the increase in consumption demand spurred by a cut in taxes would lead to an increase in output. Else, it will lead to price rise instead of increase in output.

2.3.4 Open Economy Multiplier

In an open economy there is an additional source of demand for domestic output, viz., exports (X). Exports are autonomous as they depend on foreign incomes (i.e., incomes of foreign residents who buy domestic products) rather than domestic income. Therefore, an increase in exports will have multiplier effect on income, via various rounds of induced increases in consumption. Firms and households purchase goods and services from foreign countries in an open economy. Such imports (M) are a leakage from the domestic expenditure and income streams. We assume that demand for imports depend on domestic income level. The import demand function is given by $M = h + mY$, where h is the autonomous component of import demand. Here, m is the *marginal propensity to import* (MPM) and $m > 0$. Note that both exports and imports are influenced by exchange rate changes also (we will take up such issues later in the course on International Economics). In the present Unit we consider any change in exports and imports due to exchange rate fluctuation is treated as an autonomous change.

In an open economy, exports are an injection while imports are a leakage from the stream of aggregate demand for domestic goods. Thus, aggregate demand is given by: $AD = C + \bar{I} + G + X - M = a + bY + \bar{I} + G + X - h - mY$. Therefore, the equilibrium condition is given by

$$Y = a + bY + \bar{I} + G + X - h - mY \qquad \ldots (2.9)$$

The equilibrium level of output or income is given by

$$Y_e = (a - h + \bar{I} + G + X) / (1 - b + m) \qquad \ldots (2.10)$$

In (2.10), the autonomous components are I, G and X. Any change in these autonomous components will have a multiplier effect. The autonomous expenditure multiplier for investment, government expenditure or exports is given by: $\Delta Y / \Delta G = \Delta Y / \Delta I = \Delta Y / \Delta X = 1 / (1 - b + m)$. Clearly, the open economy multiplier is smaller compared to the one for the closed economy (1/(1 – b). The reason is that a proportion m out of the increment in income is now spent on imports and domestic demand is lower to that extent as compared to a closed economy.

In general, from our discussion on multipliers you should note the following points: (i) the multiplier would be larger, the larger is the MPC (b) and MPI (d), the lower the tax rate (t) and the smaller the m; (ii) the multiplier is applicable both in case of a rise as well as a fall in autonomous incomes, meaning that income would fall by a multiple in case of a given cut in any autonomous component of demand; and (iii) the operation of the multiplier is driven by the assumption that the economy has ample excess capacity and unemployed resources, so that output can be demand-determined.

2.3.5 Limitations of the Keynesian Model

In the beginning of this section we mentioned that the Keynesian model is more relevant for an economy facing recessionary conditions, having excess production capacity in place. In economies having supply constraints (i.e., where there are supply bottlenecks), an increase in demand is likely to result in a rise in prices rather than in output. Second, Keynesian model would not apply to economies that are operating near full employment, where increase in output would require increase in factor and goods prices. Third, Keynesian model is generally used for macroeconomic analysis pertaining to the 'short run'. Normally, once wage contracts are written they last for some time; also, prices printed on sales catalogues are expected to be valid for some time. So the assumption of wage and price rigidity is quite reasonable for the short run.

Check Your Progress 2

1. State the assumptions on which the Keynesian model is based.

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2. You are given the following information for a closed economy: C = 10 + 0.75 Y and $\overline{I} = 20$.

   a) Find the aggregate demand function and calculate equilibrium income for this economy.

   b) Suppose autonomous investment increases to 25. How would equilibrium income change?

   c) What is the value of the multiplier?

   d) How would the value of the multiplier in (b) change if we introducethe government sector which imposes (i) only a lump sum tax, and (ii) only a proportional tax where $t = 0.1$?

   e) How would the value of the multiplier in (b) change if the economy is open and MPM = 0.3?

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3. By using appropriate diagram of the Keynesian model, explain how situations of disequilibrium (excess demand or excess supply) are self-correcting.

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2.4 AN ALTERNATIVE VIEW OF EQUILIBRIUM

The Keynesian model can alternatively be described through the saving-investment relationship. When saving is equal to planned investment, the economy is operating at the equilibrium level of output.

2.4.1 Saving Function

The saving function is a mirror image of the consumption function that gives the relationship between planned saving (S) and income (Y) in the economy. In the two-sector model, with only households and firms, aggregate income is either consumed or saved. Thus,

$$Y = C + S \qquad \ldots (2.11)$$

The decision to consume more is essentially the same as the decision to save less.

The saving function may be derived as follows:

$$S = Y - (a + b\,Y) = -\,a + (1 - b)\,Y \qquad \ldots (2.12).$$

Note that the intercept of the saving function is negative, indicating that 'dis-saving' (running down past saving) occur at relatively low levels of income. The slope of the saving function is $(1-b)$, which is known as the *marginal propensity to save* (MPS). Note that MPS = (1–MPC), so that higher the MPC, lower is the MPS. In economic analysis it is often assumed that richer households have higher MPS compared poor households.

In Fig. 2.6 we derive the saving function diagrammatically. The top panel represents the consumption function C, while the lower panel represents the saving function.

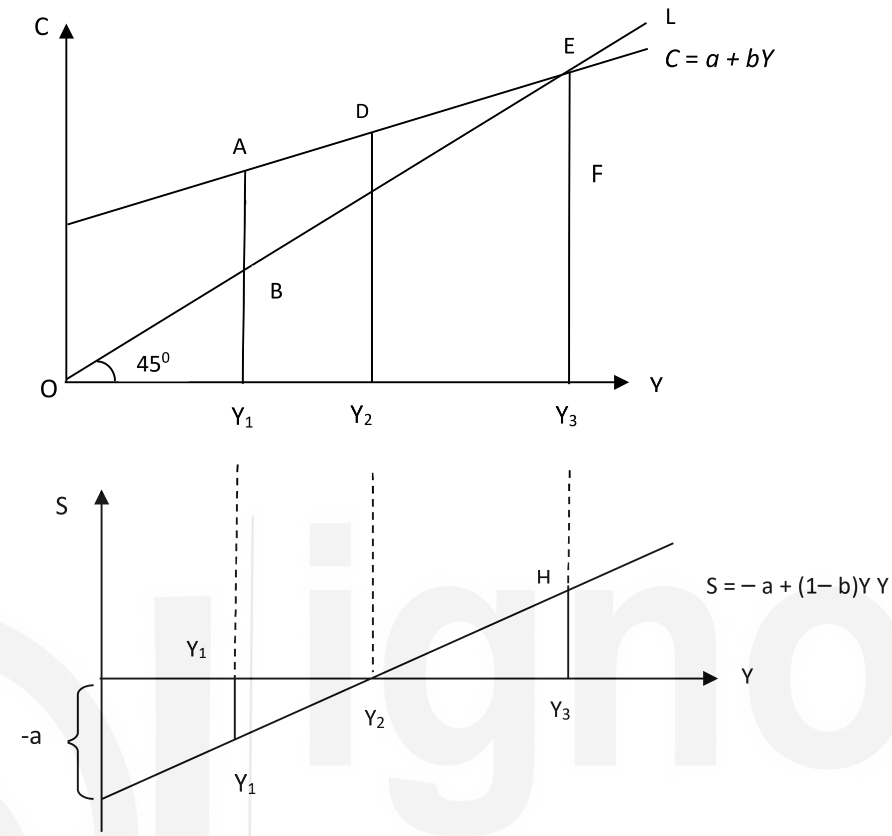

Fig. 2.6: Saving Function

At income level Y_2, where C intersects OL (the 45^0 line), planned consumption DY_2 is equal to income OY_2 (= DY_2) and planned saving is zero. At this level of income, the saving function intersects the x-axis. For all income levels less than OY_2 planned consumption exceeds income so that saving is negative. For example, in the top panel, at income level OY_1, planned consumption AY_1 is greater than income BY_1 (= OY_1) and the difference is reflected in negative saving Y_1G in the lower panel. For income levels such as OY_3, which is higher than OY_2, planned saving is positive (HY_3, in the lower panel).

2.4.2 Determination of Equilibrium Output

In a two-sector economy with only households and firms, equilibrium in the output market occurs at the level of income (or output) at which planned saving is equal to planned investment. This can be shown as follows: Suppose investment is autonomous and the investment function is given by $I = \bar{I}$ as in (2.2a). Recall that $Y = C + \bar{I}$ at equilibrium. We know that income is either consumed or saved, i.e., $Y = C + S$. Thus, we can put these together to write

$$C + S = Y = C + \bar{I}, \text{ or, } S = \bar{I} \qquad \dots (2.13)$$

In a closed economy without government, at equilibrium, planned saving must be equal to planned investment. Abstaining from current consumption (i.e., saving) involves producing fewer consumer goods and freeing resources for the production of investment goods. This can also be represented diagrammatically as in Fig. 2.7, where equilibrium income Y_e is attained at the intersection of the saving function and the investment function.

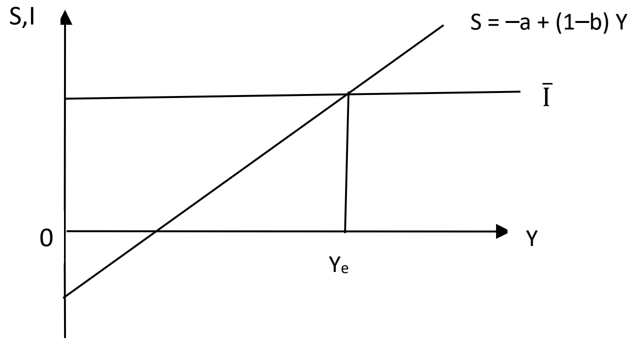

Fig. 2.7: Saving-Investment Equality

In a three-sector economy, the output market equilibrium condition is $Y = C + \bar{I} + G$. Income has three uses: consumption, saving and payment of taxes. Thus, $Y = C + S + T$. Putting these together, we can write the output market equilibrium condition as:

$$C + S + T = Y = C + \bar{I} + G, \text{ or, } S + T = \bar{I} + G \qquad \dots (2.14)$$

This can also be written as $S + (T - G) = \bar{I}$, which shows that at equilibrium in a closed economy, planned investment must be equal to total saving that include planned saving by households and public saving $(T - G)$. We can also re-write (2.14) as, $(S - \bar{I}) = (G - T)$. An implication of the above is: when planned investment is less than planned saving $(S - \bar{I} > 0)$, the government is running a budget deficit $(G - T > 0)$. The financing of the budget deficit is by borrowing from households.

2.4.3 Paradox of Thrift

An important result regarding the role of saving in the Keynesian model is referred to as the 'saving paradox' or the 'paradox of thrift'. You know that saving on the part of a household is considered as a virtue. Higher saving increases the assets and wealth of a household in the long run. For the economy as a whole, however, an increase in saving may not be good; it may be a vice. When households plan to save more, aggregate saving either remain the same or may even decrease. Let us explain it through a diagram (see Fig. 2.8).

In Fig. 2.8, equilibrium in the economy is at point A, where $S = I$. Output produced is Y_0. Suppose there is an increase in the autonomous component of households' planned saving, i.e., at each level of income planned saving is higher. This involves an upward shift of the saving function from S_0 to S_1. Note that S_1 is parallel to S_0 since there is no change in MPS. Now, the equilibrium output level increases to

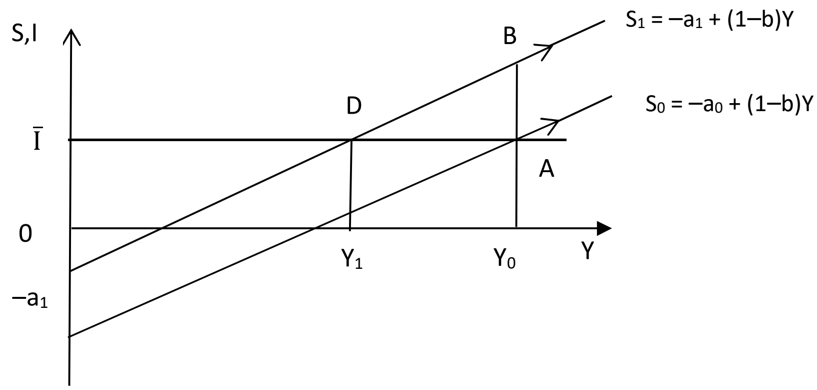

Fig. 2.8: Paradox of Thrift

Since there is no change in planned investment, there is disequilibrium in the economy now. Planned investment (DY_1) is the *same as before* ($DY_1 = AY_0$). Since people now wish to save more (saving function has shifted to S_1), income must be lower, so that saving is the same as before. The paradox is, despite an increase in thriftiness, there is no change in aggregate saving.

The desire to save more reduces planned consumption, thereby lowering aggregate demand and hence output (and income). At lower income levels, saving is correspondingly lower – instead of BY_0, aggregate saving fall to DY_1, since income has fallen to OY_1.

If the investment function has an induced component and is given by $I = c + d\,Y$, as in (2.2b), then an increase in planned saving can actually lead to lower level of saving at equilibrium, as shown in Fig. 2.9 below.

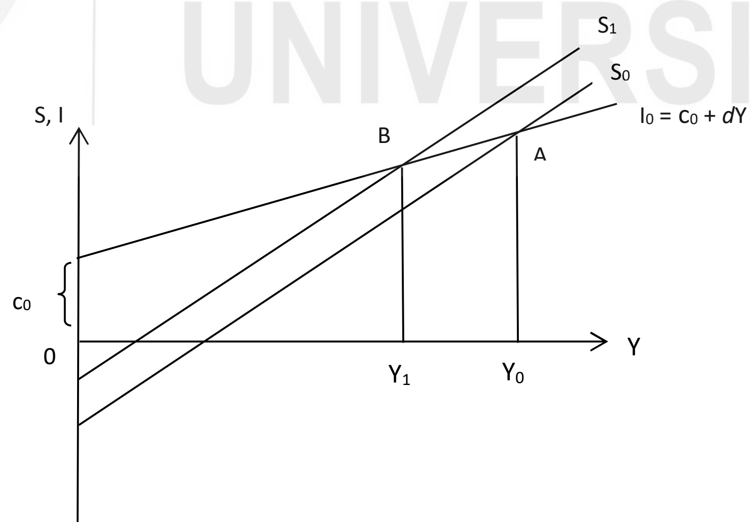

Fig. 2.9: Paradox of Thrift (Induced Investment)

In Fig. 2.9 the saving function shifts from S_0 to S_1. In this case, the reduction in consumption demand (due to rise in planned saving), leads to lower output which further leads to a decrease in induced *s*, thereby lowering income further. At the lower level of equilibrium income ($OY_1 < OY_0$), planned saving is correspondingly lower ($BY_1 < AY_0$), with the desire to save more leading to lower aggregate saving at equilibrium.

This result provides an important insight into the role of saving in the short run. In an economy that faces no supply constraints and has excess production capacity, an increase in the desire to save essentially lowers output by lowering aggregate demand. Therefore, at a lower equilibrium income level aggregate saving is the same or may even be lower than before. Note that such decline in output due to higher saving could be a short run phenomenon. In the long run, higher saving allows for higher investment and hence for creation of additional productive capacity. You will learn more about the role of saving in the long run while studying economic growth.

Check Your Progress 3

1. (a) In a two-sector economy (with only households and firms) derive the output market equilibrium condition in terms of planned saving and planned investment.

   (b) In a three-sector economy (with households, firms and government), what would be the relation between saving and investment at equilibrium if the government runs a budget surplus?

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2. What is meant by the 'paradox of thrift'? Use appropriate diagram to explain the cases of (i) investment demand is autonomous; and (ii) investment demand has an induced component.

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3. In a two-sector economy (with only households and firms), the consumption function is given by $C = 10 + 0.75\ Y$ and $\bar{I} = 20$.

   (i) Derive the saving function.

   (ii) Derive equilibrium output in terms of the equality of planned saving and investment.

   (iii) Suppose there is an increase in the desire to save while planned investment remains unchanged. The new saving function is given by $S = -8 + 0.25\ Y$. Find out the new equilibrium income level.

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2.5 LIQUIDITY PREFERENCE

To complete our understanding of the Keynesian model we have to discuss the concept of 'liquidity preference' or the demand for money in this system. An important contribution of Keynes was to point out the interest sensitivity of the demand for money. Note that demand for money refers to the demand for 'real balances', i.e., nominal money demand deflated by the general price level. Since prices are constant in the Keynesian system, any change in money demand is essentially a change in the demand for real balances.

2.5.1 Components of Demand for Money

There are three main reasons people wish to hold money. The main components of the demand for money are (i) transactions demand, (ii) precautionary demand, and (iii) speculative demand. Let us discuss each of the above.

Transaction Demand: People hold cash since it is a medium of exchange and is useful for making transactions. This gives rise to transactions demand for money. You would have noticed that people with higher income tend to hold more money for transactions compared to the poor. So it is assumed that the amount of liquid cash people wish to hold for transactions is directly proportional to their incomes.

Precautionary Demand: People wish to hold money as it would be useful in contingencies such as medical emergency, accident, etc. It is reasonable to assume that precautionary demand for money is also an increasing function of income.

Speculative Demand: According to Keynes, people also hold money for speculation in the asset market, which gives rise to speculative demand for money. To understand why speculative demand for money is sensitive to the rate of interest, consider an economy where there are only two types of financial assets – money and bonds. Money is useful for transactions, but it yields no returns, while bonds offer interest income and possibility of making capital gains when bond prices rise. There is an inverse relation between the price of bonds and rate of interest. Now consider the effect of change in interest rate on speculative demand for money. When interest rate is high, people will mostly expect it to fall and expect bond prices to rise. Therefore, they would reduce money holdings and buy bonds in the expectation of making a capital gain. So, at high rates of interest, demand for money would be low. Conversely, at low rates of interest, people expect interest rates to rise (i.e., fall in bond prices). Capital loss would induce people to sell bonds and increase the demand for money. Therefore, speculative demand for money is *inversely* related to the rate of interest.

In general, the higher the market rate of interest, the higher is the opportunity cost (interest income forgone) of holding money. So when interest rates rise, people would reduce money demand and try to hold interest-bearing assets instead which explains the inverse relation between interest rates and the demand for money. This is another way of understanding the inverse relation between money demand and interest rates, without assuming the presence of speculative demand.

From our discussion above, you should see that the money demand function can be expressed as follows:

$$L = q\,Y - t.i \qquad \ldots (2.15)$$

where $q > 0$, $t > 0$; L is the demand for real balances, Y is aggregate income and i is the rate of interest. Diagrammatically the money demand function can be shown as a downward sloping line in Fig. 2.10, where interest rate is presented along the vertical axis and money demand is along the horizontal axis.

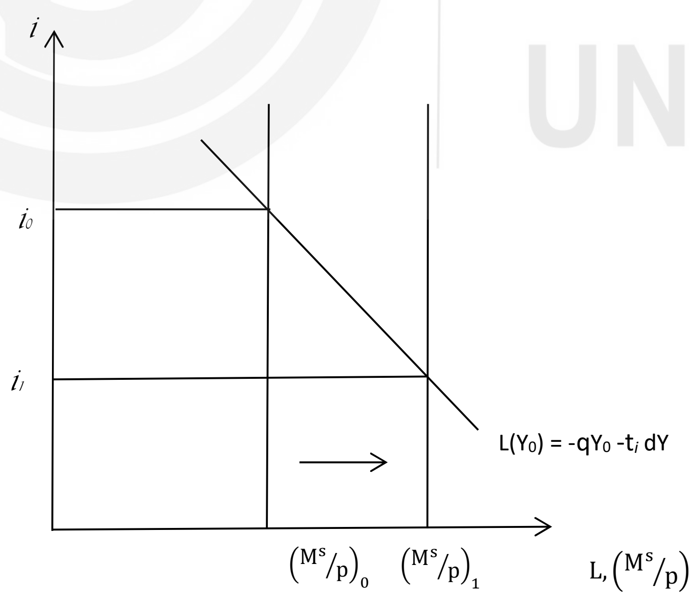

Fig. 2.10: Demand for Money

Note that the money demand function $L(Y_0)$ represents demand for money at the income level Y_0. Movement along $L(Y_0)$ demonstrates the interest sensitivity of the money demand function. As interest rate falls from i_0 to i_1, there is an increase in the demand for money due to speculative demand for money, as discussed above. For all income levels, demand for money greater than Y_0 would be higher at each interest rate (since transactions and precautionary demands for money increase with income), and the money demand curve would shift to the right. For incomes lower than Y_0, money demand would be lower at each interest rate and the money demand curve would shift to the left.

Money supply is determined by the central bank of the country (you will learn more about it in subsequent Units) and is given by the vertical line M^s/P, where M^s is nominal money supply, P is the price level and M^s/P is the real money supply. Equilibrium in the money market occurs at the intersection of money supply and money demand. Note that with the downward sloping money demand function, an increase in money supply (from $(M^s/P)_0$ to $(M^s/P)_1$) brings about a reduction in the rate of interest (from i_0 to i_1).

2.5.2 Liquidity Trap

A special situation can arise if interest rates become unusually low and hit the lowest possible level, so that the demand for money becomes infinitely elastic and the economy faces a 'liquidity trap'. This can happen when interest rates have fallen so low (and bond prices have risen so high) that everyone expects interest rates to rise (and bond prices to fall). Therefore no one wants to hold bonds, in fear of making a capital loss and everyone wants to hold money. In this situation, the money demand curve becomes horizontal as presented in Fig. 2.11 and *t* in equation (2.16) tends to infinity.

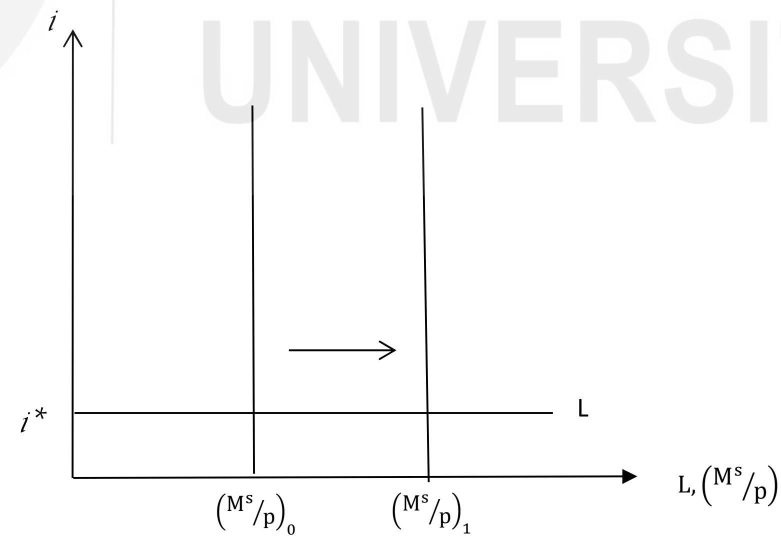

Fig. 2.11: Liquidity Trap

When interest rates hit a floor such as i^*, any injection of extra liquidity in the system is absorbed, as people willingly hold money (liquidity) rather than bonds at an unchanged rate of interest, i^*. Therefore once the economy is in a liquidity trap monetary policy cannot lower the rate of interest beyond i^* and hence it cannot affect real variables like s. An increase in money supply (from $(M^s/P)_0$ to$(M^s/P)_1$) has no impact on the interest rate which remains unchanged at i^*. Once you learn about monetary and fiscal policies, you will see that conventional monetary policy becomes ineffective when the economy is in a liquidity trap whereas fiscal policy is very effective in this case.

The concept of liquidity trap became important in policy circles during the global financial crisis of 2007-08, when in advanced economies like USA and Japan interest rates hit a floor and became zero (USA) or almost zero (Japan). In this case conventional monetary policy could not be used to reduce the rate of interest on government bonds any further and policy experiments involving the use of unconventional monetary policy measures had to be explored. Fiscal policy measures were also employed, but increase in public debt levels raised concerns regarding debt sustainability. In the next section we will explore the role of fiscal policy in the Keynesian model and touch on some of these policy issues.

2.6 ROLE OF GOVERNMENT

The classical economists, as we saw in Unit 1, advocated minimal government intervention in determination of prices and wage rate. According to them the activity of the government should be limited to law and order, justice and defence. They assumed that prices and wage rate are flexible, and there is perfect competition in the markets. Consequently, there is no unemployment in the economy.

The Keynesian economics is in sharp contrast to the classical economics. Keynesian model implies that the government has an important role to play in the economy. The government should play an active role to bring in equilibrium in the market. Keynes highlighted that rigidities in prices and wage rate prevent smooth functioning of market mechanism in the real world. He also highlighted the critical role of effective demand in determining the level of output and employment in a macroeconomic context. The classical economists had focused almost exclusively on the supply side (i.e., on production of output). In contrast, Keynesian analysis showed that aggregate demand (i.e., planned spending by economic agents) plays a key role in determining the level of output and employment in the short run.

Keynes pointed out that when lack of effective demand is the underlying cause of an economic recession. When there is high unemployment, there is tendency for wage rate to decline. This will worsen the situation by reducing purchasing power and hence effective demand. Therefore, government should intervene by increasing public expenditure; it should go for a deficit budget. Fiscal and monetary policies can be used to inject additional demand in the economy, so as to enhance spending (aggregate demand). That would lead to increases in output, income and employment in the economy, and the economy would emerge out of the recession.

The success of Keynesian policies during the Great Depression (1929-33) made a strong case for an activist role of the government. In particular, Keynesian economists advocated the use of discretionary fiscal policies, as Keynes had pointed out the limitations of monetary policy. If the economy was in a liquidity trap, conventional monetary policy would be completely ineffective, while fiscal policy would still remain effective.

During the global financial crisis of 2007-08, most countries used the Keynesian prescription of expansionary fiscal policy (increases in government spending) to boost economic activity, although it resulted in widening budget deficits and high levels of public debt. High interest burden on public debt contributes to further widening of budget deficits, raising concerns about sustainability of the deficits. Further, high interest rate may lead to 'crowing out' of private investment.

You must be aware that during 2020 and 2021, the world economy suffered a severe economic crisis due to Covid-19. There was acute shortage in aggregate demand due to decrease in economic activity. The response of the governments to this crisis situation brought to the fore the importance of the policies advocated by Keynes.

Check Your Progress 4

1. Discuss some of the factors that influence the demand for money.

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2. What is speculative demand for money? How does it affect the slope of the money demand function?

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3. What is meant by 'liquidity trap'? What are its implications?

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4. Discuss the role of the government in an economy that is facing a prolonged recession.

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5. Give any three reasons why policy makers tend to be concerned about mounting budget deficits.

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2.7 LET US SUM UP

Keynesian model is appropriate for an economy facing recessionary conditions with unemployed resources and lack of effective demand. Keynes assumes that markets are imperfect with wage and price rigidity. Within this framework, output is demand-determined. An increase in any autonomous component of aggregate demand has a multiplier effect, leading to several rounds of increment in output. Our discussion on the consumption and investment functions highlighted the factors that influence these two important components of aggregate demand.

Equilibrium in the Keynesian model can be viewed in terms of the equality between planned saving and planned investment. We learnt about the paradox of thrift, which shows that an increase in the propensity to save results in a situation where aggregate saving may remain unchanged or decline. Since output is demand-determined in this model, an increase in planned saving leads to lower aggregate demand and hence lower output, explaining why we observe the paradox of thrift.

The demand for money (liquidity preference) increases with increase in income and decreases with increase in interest rate. The inverse relationship between money demand and the rate of interest could be explained by the Keynesian concept of speculative demand for money. We also learnt about the liquidity trap – at unusually low rate of interest the demand for money becomes infinitely elastic and traditional monetary policy became ineffective.

2.8 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) Consumption function is a linear function given by $C = a + bY$. The APC declines with increase in Y while the MPC remains constant.

2) i) A decrease in autonomous consumption will lead to a parallel downward shift in the consumption function.

   (ii) An increase in MPC will increase the slope of the consumption function.

3) Autonomous investment remains constant irrespective of the level of income or interest. Induced invest depends on income or rate or interest or both. See Sub-Section 2.2.2 for details.

4) As interest rate decreases the cost of borrowing decreases. Thus, some of the projects which were unviable earlier become profitable. Therefore, more investment takes place with decrease in interest rate.

Check Your Progress 2

1) Keynesian model is based on the assumption that resources are under-utlised, prices and wage rate are rigid, and output is a homogenous commodity.

2) a) Aggregate demand function is: $Y = 30 + 0.75Y$. Equilibrium income is 120.

   b) Equilibrium output will increase to 140.

   c) 4

   d) In the case of lump sum tax, the multiplier will be the same as autonomous expenditure multiplier given in (b). In the case of proportional tax at the rate of 0.1, the multiplier will be reduce to $\frac{1}{1-0.75\times0.9} = 3.08$

   e) The open economy multiplier is given by *1 / (1 – b + m*). Therefore, the multiplier will be 1.8.

3) Go through Sub-Section 2.3.1 and answer.

Check Your Progress 3

1) (a) Go through Sub-Section 2.4.2; explain Fig. 2.7.

(b) Saving will be lower than investment, since (S + T) = ($\bar{I}$+G).

2) Refer to Fig. 2.8 and Fig. 2.9 for the answer.

3) (i) $S = -10 + 0.25Y$; (ii) 120; (iii) 112

Check Your Progress 4

1) You should discuss about transaction demand, precautionary demand and speculative demand for money.

2) Refer to Sub-Section 2.5.1 and answer.

3) It refers to the insensitiveness of the demand for money to rate of interest, when the rate of interest is very low. Conventional instruments of monetary policy become ineffective in liquidity trap.

4) Government should increase public expenditure so that there is no decrease in effective demand.

5) High fiscal deficit increases government debt burden, increases interest rate amd may lead to crowding out of private expenditure.

UNIT 3 THE NEOCLASSICAL SYNTHESIS*

Structure

3.0 OBJECTIVES

After going through this unit, you should be in a position to

- explain the equilibrium in the real sector and the monetary sector in an economy;
- explain the underlying ideas behind the IS curve;
- explain the underlying ideas behind the LM curve;
- explain how the IS and LM curves interact;
- identify factors that influence the position and slope of the IS and the LM curves;
- derive aggregate demand (AD) curve from the IS-LM model; and
- find out the factors that influence the AD curve.

3.1 INTRODUCTION

In Unit 2 we discussed how the equilibrium level of output is determined in the Keynesian model. Recall that Keynes came up with his analysis in the aftermath of the Great Depression. He considered an economy with underutilization of resources (i.e., presence of idle capacity) with an objective of correcting the

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economy immediately. Thus, he focused on the short-run. He reasoned that an increase in aggregate expenditure (which represents aggregate demand) leads to an increase in output; it does not lead to rise in prices till full employment is reached. In view of above, price level is kept fixed in the simple Keynesian model. An implication of fixed prices is that nominal values and real values are the same.

In this Unit we deal with a closed economy; there is no external trade. We begin with the derivation of IS and LM curves. Subsequently, we derive the AD curve from the equilibrium points of the IS-LM model.

3.2 EQUILIBRIUM IN THE REAL SECTOR

The aggregate expenditure (or, aggregate demand) of an economy is given by

Y = C + I + G

where Y is output (or, aggregate supply), C is consumption expenditure, I is investment expenditure, and G is government expenditure.

The consumption function is given by

$$C = a + b(Y - T) = a + bY_d \qquad \dots (3.1)$$

where Y_d is disposable income (i.e., Y – T). Here T is a lump sum tax levied by the government. The investment function is given by

$$I = c + dY - e\,i \qquad \dots (3.2)$$

where i is the rate of interest (there is no difference between nominal rate of interest and real rate of interest as price is kept fixed). Let us assume that government expenditure (G) is constant, and exogenously given. Thus,

$$Y = a + b(Y - \bar{T}) + (c + dY - e\,i) + \bar{G} \qquad \dots (3.3)$$

On re-arrangement of terms in equation (3.3), we obtain

$$Y - bY - dY = a - b\bar{T} + c - ei + \bar{G}$$

Or, $Y = \frac{a+c+\bar{G}-b\bar{T}}{(1-b-d)} - \frac{e}{(1-b-d)}i$ $\qquad \dots (3.4)$

The IS equation, in its general form, is given by

$$Y = \alpha_G(\bar{A} - ei) \qquad \dots (3.4a)$$

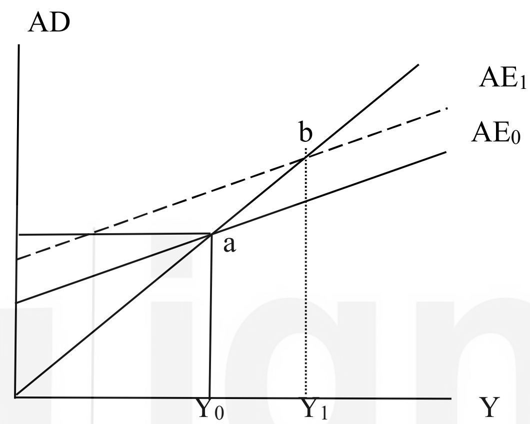
where $\alpha_G = \frac{1}{(1-b-d)}$ and $\bar{A} = a + c + \bar{G} - b\bar{T}$.

In equation (3.4) we find an inverse relationship between Y and i. It implies that, as i increases, there is a decrease in Y. In order to find out the slope of equation (3.4) you have to express it in terms i and find out the slope $\frac{di}{dY}$. You can find out the slope of the IS curve as $-\frac{(1-b-d)}{e}$. Depending upon the value of the parameters, the IS curve would be flatter or steeper.

3.2.1 Derivation of the IS Curve

In Fig. 3.1 we explain the relationship between Y and i through a diagram. In Panel (a) of Fig. 3.1 we take aggregate expenditure (AE) on the y-axis and output (Y) on the x-axis. Let us assume that the economy is operating with AE_0 while the rate of interest is i_0. The intersection of AE_0 with the 45^0 line at point 'a' is of utmost importance to us. Note that the 45^0 line in Fig. 3.1(a) indicates equality between aggregate demand (AE) and aggregate supply (Y), such that it corresponds to equilibrium in the economy. We plot this point 'a' in panel (b) of Fig. 3.1. In panel (b) we take interest rate (i) on the y-axis and output (Y) on the x-axis.

Panel (a)



Panel (b)

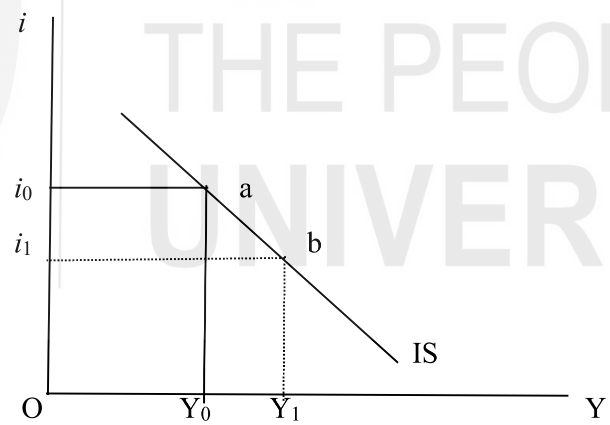


Fig. 3.1: Derivation of the IS Curve

Suppose, there is a decrease in the rate of interest from i_0 to i_1. As a result of this, the level of investment in the economy increases and the new aggregate expenditure level rises to AE_1. The aggregate expenditure curve (AE_1) intersects the 45^0 line at point-b. Due to the increase in the aggregate expenditure, there is an increase in the equilibrium level of output from Y_0 to Y_1. We plot point-b in Fig. 3.1(b) corresponding to output level Y_1 and intertest rate i_1. When we combine points 'a' and 'b' we obtain a downward-sloping curve, called the IS curve. A point on the IS curve denotes equilibrium in the real sector (also called the goods sector) of the economy and there is equality between investment (I) and saving (S). Note that the real sector of the economy is at equilibrium on each and every point of the

IS curve. An implication of the above is that any point outside the IS curve shows disequilibrium in the real sector. In Fig. 3.2 we consider two such points 'c' and 'd' which are not on the IS curve, and find out their implications.

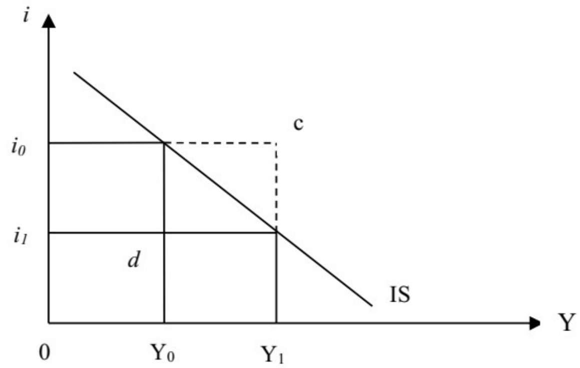


Fig. 3.2: Disequilibrium in the Real Sector

At point 'c' in Fig. 3.2, interest rate is i_0 while output is Y_1 . Notice that at Y_1 level of output, the equilibrium rate of interest should be i_1. Thus, the actual rate interest is higher than what it should. As a result, the aggregate demand will decrease (largely due to decrease in investment) and the rate of interest will decline. It implies that at point-c, there is excess supply of goods in the market. We can generalize that any point which is above and to the right of the IS curve, indicates excess supply of goods.

Let us look at point 'd' which is below and to the left of the IS curve. At point 'd' in Fig. 3.2, income is at Y_0 but the rate of interest is at i_1. Since the rate of interest is lower than the equilibrium rate of interest, the demand for investment will increase. Consequently, the rate of interest will rise upward. Thus, at point 'd' there is excess demand for goods. We can generalize that at any point below and to the left of the IS curve, there is excess demand for goods.

3.2.2 Shift in the IS Curve

The IS curve depicts the relationship between income and interest rate. Therefore, any change in other parameters or variables will result in a shift in the IS curve. The position and the slope of the IS curve depends upon the value of the parameters in equation (3.4). If the IS curve is flatter, the economy is more sensitive to interest rate – a small decrease in interest rate will lead to a large increase in income. On the other hand, a steeper IS curve implies insensitivity to interest rate change.

There are two exogenous variables in equation (3.4), viz., G and T. In equation (3.4) we assumed that government expenditure and taxes are fixed. Thus, changes in taxes and government expenditure will also result in shifts in the IS curve. Let us look into equation (3.4). An increase in the value of $\bar{G}$ will result in an increase in the intercept; thus, there will be an upward parallel shift in the IS curve. An increase in $\bar{T}$, on the other hand, will decrease the intercept and shift the IS curve downward to the left.

A change in the value of a parameter leads to a shift in the IS curve. An increase in autonomous consumption (the coefficient 'a' in equation (3.4)) or autonomous investment (the coefficient 'c' in equation (3.4)) will lead to a parallel shift of the IS curve upward to the right. A change in the marginal propensity to consume, however, will affect the slope of the IS curve.

Check Your Progress 1

1) What will be the nature of change in the IS curve if there is a decrease in the propensity to consume?

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2) What is the implication of the following on the position of the IS curve?

(i) Decrease in autonomous government expenditure

(ii) Decrease in tax rate

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3.3 EQUILIBRIUM IN THE MONETARY SECTOR

The derivation of LM curve utilises the Keynesian view that speculative demand and transaction demand for money are separate. The speculative demand for money depends on the rate of interest while the transaction demand and the precautionary demand for money depend on the level of income. Thus, total demand for money (M) depends on both income (Y) and rate of interest (i). The demand for money in real terms ($\frac{M^d}{P}$) is given by

$$L = kY - hi \qquad \dots (3.5)$$

where $k > 0$, $h > 0$; L is the demand for real balances (liquidity), Y is aggregate income and i is the rate of interest. You should remember three issues about the monetary sector: (i) A person would like to deposit his money in bank and earn interest on it. The alternative is to hold on to the cash, where no interest income is earned. Thus, the opportunity cost of holding money is the interest foregone. If there is an increase in the rate of interest, the interest foregone increases. Conversely, if the rate of interest decreases, the interest income foregone decreases. Therefore, the demand for money is a downward-sloping curve, if we measure rate of interest on the y-axis and quantity of money on the x-axis. You may be familiar with the concept of liquidity trap; a situation where the rate of interest is very low, and people prefer to keep all their money in the form of liquid cash as the interest income foregone is negligible. (ii) The demand for real balances is described for a given level of income. We consider the money demand function while keeping the level of income fixed (say, $L(Y_0)$). If there is an increase in the level of income, there is an increase in the level of transactions and wealth in the economy, which in turn increases the demand for money. Thus, there will be a parallel shift in the money demand function. (iii) The supply of money is determined by the central bank and it is constant. Thus, the supply of money is a vertical straight line. If the supply of money is increased by the central bank, the vertical straight line will shift to the right. When we deal with the short-run, we assume that money supply does not change.

3.3.1 Derivation of the LM Curve

The supply of money in real terms is given by $\frac{M^s}{P}$. Equilibrium in the money market is achieved when $L = M = \bar{M}$. Thus, equilibrium in the money market is given by

$$\frac{\bar{M}}{\bar{P}} = kY - hi \quad \ldots (3.6)$$

We can re-arrange terms in the above equation and obtain

$$Y = -\frac{1}{h}\frac{\bar{M}}{\bar{P}} + \frac{h}{k}i \quad \ldots (3.6a)$$

There is a positive relationship between Y and i in the LM curve. In order to find out the slope of equation (3.6) you have to express the equation in terms i and find out the slope $\frac{di}{dY}$. You can find out the slope of the LM curve as $\frac{k}{h}$. Depending upon the value of the parameters, the LM curve would be flatter or steeper.

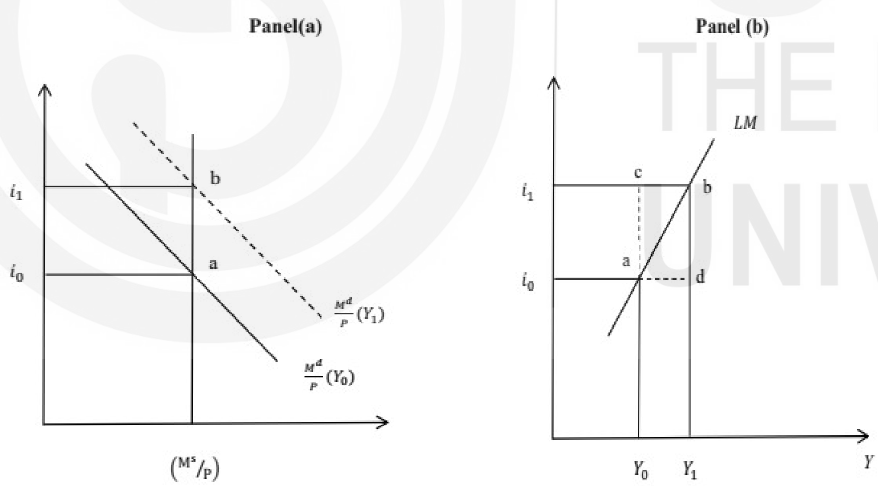

Fig. 3.3: Derivation of the LM Curve

In Fig. 3.3, Panel (a) we depict the equilibrium in the money market (or, in the monetary sector). We take demand for real balances ($\frac{M}{P}$) on the x-axis and the rate of interest on the y-axis. Let the money demand in the economy be given as $\frac{M^d}{P}(Y_0)$ when the aggregate income is Y_0. In panel (a) of Fig. 3.3, this equilibrium is represented by point 'a'. The equilibrium rate of interest and income are i_0 and Y_0 w respectively.

Let there be an increase in the level of income from Y_0 to Y_1. Such an increase in income will lead to a parallel upward shift in the money demand function. We depict this money demand function as a dotted line $\frac{M^d}{P}(Y_1)$ in Panel (a) of Fig. 3.3. With money supply remaining constant, an increase in aggregate demand will lead to a rise in the interest rate. The economy is now at equilibrium with output level Y_1 and interest rate i_1. The intersection between the money supply function and the money demand function is denoted by point 'b' in Fig. 3.3, panel (a).

In panel (b) of Fig. 3.3 we plot the LM curve by taking combinations of interest rate and aggregate income where the money market is in equilibrium. We have two such points, 'a' and 'b' in panel (a) of Fig. 3.3. You should note that the LM curve is the combination of all the points where the money market is in equilibrium. The LM curve is upward-sloping as you can see from the figure. An implication of the LM curve is that as income increases people want to hold more money, thus driving up the interest rate. If the speculative demand for money is perfectly elastic to interest rate (liquidity trap situation) the LM curve will be horizontal. On the other hand, if speculative demand is perfectly inelastic to interest rate, the LM curve will be vertical.

To the left side of the LM curve there is excess supply of money (for example, point 'c' in Fig. 3.3 panel (b)). Excess supply in the money market implies that the demand for money is less than its supply. It means that people want to reduce the amount of liquid cash they possess. This induces people to buy bonds so that bond prices rise and rate of interest declines. A decline in the rate of interest restores equilibrium in the economy. To the right side of the LM curve, there is excess demand for money (for example, point 'd' in Fig. 3.3 panel (b)). Excess demand for money causes people to sell bonds (so that they can keep higher amount of liquid cash). This leads to a fall in bond prices and rise in the rate of interest. This way equilibrium in the monetary sector is restored.

3.3.2 Shift in the LM Curve

The position and the slope of the LM curve depend upon certain factors, including money supply. An increase in money supply shifts the LM curve to the right. As a result, at every level of output, the equilibrium rate of interest is lower than before. Thus, increase in money supply shifts the LM curve to the right. Conversely, a decrease in money supply will shift the LM curve to the left so that equilibrium rate of interest is higher at every level of output.

3.4 SIMULTANEOUS EQUILIBRIUM OF REAL AND MONETARY SECTORS

Now let us combine IS and LM curves as shown in Fig. 3.4. Such integration of IS-LM gives a unique combination of i and Y, which represents equilibrium in both real market and money market. In Fig. 3.4 we superimpose IS_0 and LM_0 in the same diagram. We measure income (Y) on the x-axis income (Y) and rate of interest on the y-axis. The equilibrium rate of interest and the equilibrium output level are given by i_0 and Y_0 respectively.

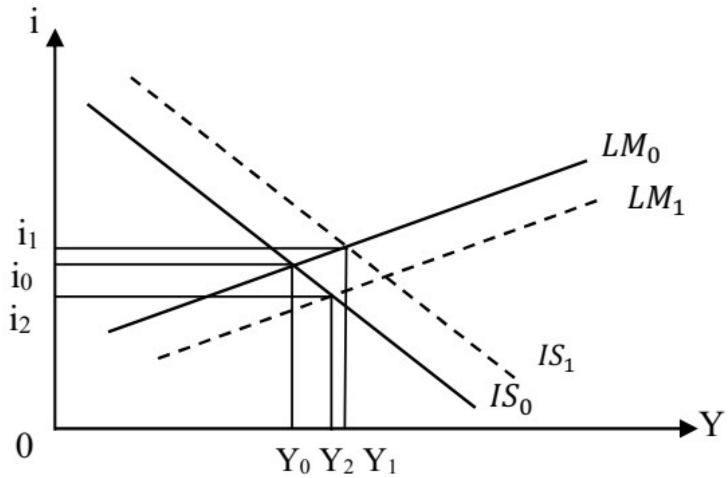

Fig. 3.4: Simultaneous Equilibrium

3.4.1 Shocks in the IS-LM Model

Previously we have discussed about the factors that lead to a shift in the IS curve and the LM curve. Often the economy experiences exogenous demand shocks and supply shocks. Let us analyse the impact of such shocks in terms of the IS-LM model. Suppose there is a boom in the stock market, which changes the wealth of households. Such a situation will increase the level of consumption in the economy. Consequently, there will be an increase in aggregate expenditure and the IS curve will shift to the right from IS_0 to IS_1 (see Fig. 3.4). As a result, there is an increase in output from Y_0 to Y_1. A negative shock such as a decrease in business confidence of firms will shift the IS curve to the left. Such a left-ward shift in the IS curve will decrease output as well as interest rate.

Let us consider another example. Suppose there is a series of online financial fraud cases in the country. This will send a negative signal to people – this will discourage people to keep their income in the formal system; there will be a decline in online transactions and the demand for money will increase. As a result, the LM curve will shift to the right from LM_0 to LM_1(see Fig. 3.4). Consequently, there will be an increase in output from Y_0 to Y_2, and decrease in interest rate from i_0 to i_2.

You should note that the nature and extent of change in output and interest rate will depend upon the slope of the IS and LM curves. If the LM curve is flatter, monetary policy is quite effective. A small decrease in interest rate by the central bank will increase output by a large amount. If the LM curve is steeper, monetary policy will not be effective. In order to increase output by a small amount we will need a large decrease in the interest rate.

3.4.2 Adjustment Process in the Economy

A question arises at this point: What happens to the economy if it is operating at a point which is not at the intersection of the IS and LM curves? Does it automatically converge to the equilibrium point? Let us try to answer this question.

If the economy is not operating at the unique point where the IS and LM curves intersect, there is disequilibrium in the economy. In that case an adjustment process starts and the economy moves to an equilibrium point. The following two issues are important in this context:

a) If there is an excess demand for money (to the right of the LM curve), the interest rate will increase. The interest rate will fall in case we observe excess supply in the money market. Fall in interest rate will lead to increase in investment. Increase in investment will affect the IS curve.

b) Excess demand for goods (to the left of the IS curve) leads to an increase in output. Increase in output will lead to increased demand for money. Increase in demand for money will lead to rise in interest rate and outward shift in the LM curve.

We observe from the above that both real sector and monetary sector are inter-related. A real shock to the economy will have effects on the monetary sector and *vice versa*.

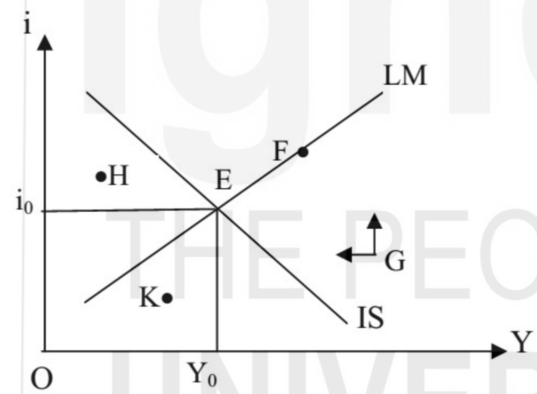

Fig. 3.5: Convergence to Equilibrium Point

In Fig. 3.5 we plot the IS and LM curves. The economy is in equilibrium at point E, with i_0 to Y_0 as equilibrium interest rate and output respectively. Suppose, due to certain shock to the economy, the equilibrium is disturbed. We have indicated the points F, G, H and K where the economy is not in equilibrium.

Let us consider point G. As mentioned earlier, at point G, there is excess demand for money and excess supply of goods. The arrows given at point G specify the direction in which adjustment takes place. Due to excess demand for money, there will be a rise in the rate of interest. This is because people will sell bonds so as to hold more of liquid cash, as a result of which the bond prices will decline and rate of interest will increase. Upward pointing arrow in Fig. 3.5 represents rising interest rate. At point G, there is excess supply of goods also. This will lead to involuntary accumulation of goods; inventories will increase. As a result, firms will cut down production and the level of output will decline. The leftward pointing arrow represents the declining output. Finally, the economy will move towards equilibrium point E.

Points F, H and K in Fig. 3.5 also are disequilibrium points. You can analyse the supply and demand conditions at these points and find out the directions in which adjustment will take place. For example, at point 'H' there is excess supply of money (since it is to the left of the LM curve) and excess demand for goods (since it is to the left of the IS curve). Excess supply of money will lead to a decrease in interest rate. Excess demand for goods will lead to an increase in output level. Consequently, the economy will move towards the equilibrium point E.

Suppose the economy is at point F (see Fig. 3.5) – here the monetary sector is in equilibrium, but the real sector is in disequilibrium. There is an excess supply of goods in the economy. This will lead to an increase in inventory accumulation. As a result, firms will decrease their production levels. Decrease in output will lead to a decrease in the demand for money and consequent decrease in the rate of interest. Finally, the economy will reach the equilibrium point E where both the IS and LM curves intersect.

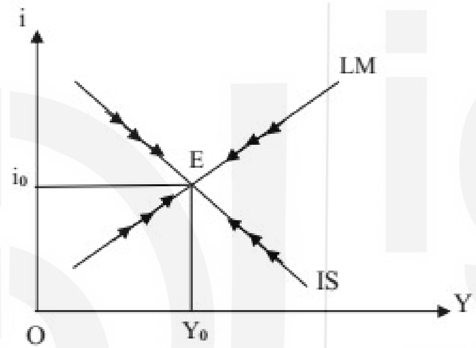

Fig. 3.6: Adjustment Process in the Economy

Note that when one of the two markets is observing disequilibrium, the economy reaches back to equilibrium after the adjustment process. In Fig. 3.6 we have indicated the direction of change by arrow marks.

3.4.3 Impact of Supply Shocks

In the IS-LM model it is not necessary that the equilibrium occurs at the level of potential output of the economy (or, full employment output). Equilibrium could be at any level of output in the economy. An objective of policy makers has always been that the economy should operate at potential output level with unemployment at its natural level. In the long run does the economy maintain equilibrium at the full employment output level?

In the long run prices do not remain constant. If actual output is above the potential out, there will be price rise. The aggregate expenditure of the economy (such as consumption, investment and government expenditure) describes the demand for goods and services, which refer to physical quantities of goods and services. A change in the price level may not affect these quantities. Thus, the IS curve is relatively stable – it may not shift. The LM curve, on the other hand, is based on the supply of and demand for real balances. Thus, a change in price level will shift the LM curve.

The full employment output is given by the production capacity of the economy. It is derived from the production function (see Unit 1). Suppose the full employment output (potential output) of the economy is given by Y^*. We depict it by a vertical straight line in Fig. 3.7. The IS_0 curve and LM_0 curve intersect at E which corresponds to the full employment output.

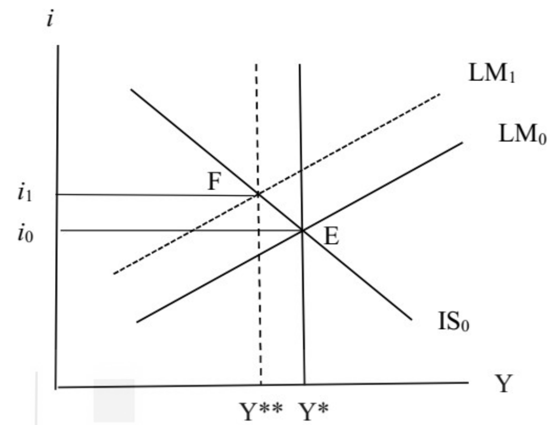

Fig. 3.7: Adjustment to Adverse Supply Shocks

Suppose the economy gets an adverse supply shock (for example, decline in agricultural productivity due to drought), as a result of which there is a decrease in the potential output of the economy from Y^* to Y^{**}. As pointed out above, the IS curve is relatively stable in such situations and the economy adjusts through a shift in the LM curve. Due to the adverse supply shock, there is a rise in prices and consequent leftward shift in the LM curve. Thus, there is a rise in the rate of interest and the new equilibrium is at F.

Check Your Progress 2

1) Explain how the LM curve is derived.

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2) Suppose the LM curve is vertical. What are its implications?

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3) In the IS-LM model, explain how adjustment to supply shocks takes place.

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3.5 AD-AS MODEL

In the IS-LM model we assumed that price level is constant. Let us relax that assumption and look into the impact of prices on equilibrium output in the economy. First, we derive the aggregate demand (AD) curve. Subsequently, we juxtapose the AD curve with the aggregate supply (AS) curve and find out the equilibrium level of output.

3.5.1 Derivation of the AD Curve

A change in the price level changes the real money supply ($\frac{M^S}{P}$) in the economy. This will result in a shift in the LM curve if money supply (M) is assumed to be constant. Thus, there will be a change in the equilibrium rate of interest and output. If we assume that the price level keeps changing over time, we will obtain a series of equilibrium levels of output corresponding to different price levels. If we plot such combinations of prices and output, we obtain the aggregate demand (AD) curve.

The derivation of the AD curve from IS-LM curves is shown in Fig. 3.8. In panel (a) of Fig. 3.8 we juxtapose IS_0 and LM_0 curves intersecting at point E with an equilibrium income level of Y_0 and interest rate i_0. The LM_0 curve is based on real money stock of M_0/P_0. Thus, we describe the combination P_0 and Y_0 as point 'a' on the *AD* curve in panel (b) of Fig.3.8.

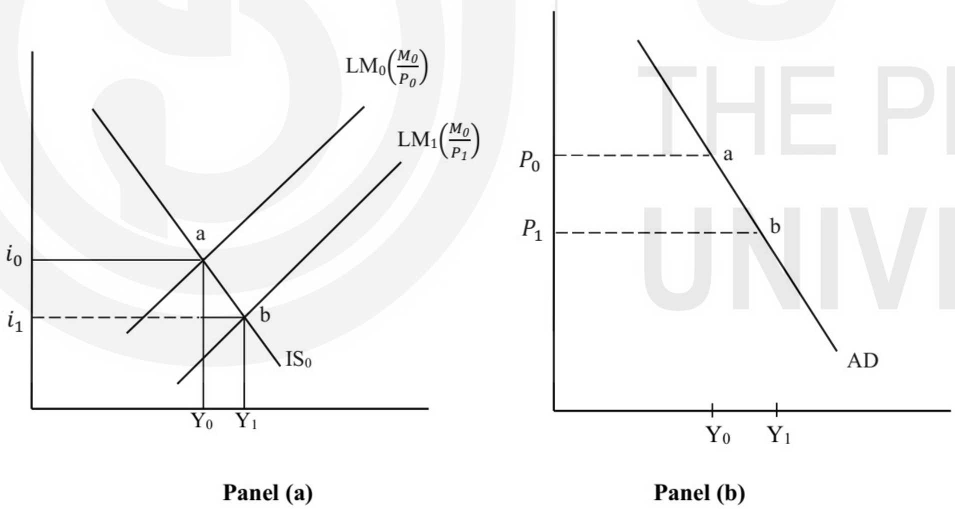

Fig. 3.8: Derivation of AD Curve

Now, let us assume that price level declines to P_1 . If there is no change in money stock, supply of real balances increases to M_0/P_1. Note that the increase in real balances is due to decrease in prices, not due to an increase nominal money stock. The *LM* curve shifts to the right. In panel (a) of Fig. 3.8 we depict the new LM curve as LM_1. With IS_0 and LM_1, the new equilibrium point is 'b'; the rate of interest falls to i_1 and income rises to Y_1. The combination of P_1 and Y_1 becomes point 'b' in Panel (b) of Fig. 3.7. By joining all such points as 'a' and 'b', we obtain the AD curve.

The AD curve is downward-sloping, which indicates an inverse relationship between prices and output. A decline in price level increases the real money supply, which creates disequilibrium in the money market. Increase in the supply of real balances lead to a decrease in interest rate. A decrease in the rate of interest restores equilibrium in the money market. A decrease in the rate of interest leads to an increase in the level of investment. An increase in the level of investment leads to an increase in aggregate expenditure, thereby increasing the level of output.

The AD curve is flatter, if a given change in price leads to a larger change in output. On the other hand, the AD curve is steeper if a given change in price leads to a smaller increase in output. The AD curve shifts upward to the right as a result of expansionary fiscal policy (e.g., increase in government expenditure, reduction in tax rate). A contractionary fiscal policy, on the other hand, shifts the AD curve downward to the left. Similarly, an expansionary monetary policy (e.g., reduction in interest rate, increase in money supply through open market operations, reserve requirements, etc.) will shift the AD curve to the right while a contractionary monetary policy will shift the AD curve to the left.

You should note that any factor that leads to a shift in the IS curve or the LM curve to the right will shift the AD curve to the right side. Similarly, any factor that shifts the IS curve or LM curve to the left will shift the AD curve to the left. Thus, an increase in any component of autonomous spending such as an increase in government spending leads to a rightward shift of the AD curve. Also, an increase in the nominal money supply will shift the AD curve upwards.

3.5.2 Aggregate Supply (AS) Curve

From microeconomics we know that the AS curve is upward-sloping – when price of a product increases, firms produce more of the product. In the case of the whole economy, production capacity is limited. Resources available to the economy are not unlimited. The behaviour of the AS curve is different in the long run compared to the short run. Short run, as you know, is a time period in which wages and certain other economic variables do not respond to changes in economic environment. In the long run is a time period in which wages and prices are flexible.

Classical theory discussed in Unit 1 assumed that the AS curve is vertical at the full employment level. Thus, output is perfectly inelastic to changes in prices. Keynes went to the other extreme and assumed that the AS curve is horizontal. An implication of the horizontal AS curve is that firms can increase production to any level without affecting prices. Both the versions (vertical and horizontal) of the AS curve seems unrealistic. As you will learn in later units, the new-classical economists suggest that the AS curve is upward-sloping in the short run. Lucas' in his 'aggregate supply hypothesis' advocate that economic agents respond to price and wage incentives, which influences their trade-off between leisure and work. As a result, when actual wage is higher than equilibrium wage rate, households supply more labour. Conversely, when actual wage rate is lower than equilibrium wage rate, household will decrease their supply of labour. As a result of this, the short run aggregate supply (SRAS) curve would be upward-sloping. The long run aggregate supply curve (LRAS), however, will be vertical. Any change in the production capacity of the economy will shift the LRAS curve.

In Fig. 3.9 we depict the AD curve and an upward-sloping AS curve. The equilibrium in the economy is described by point E corresponding to output Y_0 and prices P_0. Suppose there is a favourbale demand shock to the economy so that the AD curve shifts to AD_1. As a result, there are increases in both output and prices. The new equilibrium described by point F corresponds to output Y_1 and prices P_1. The increase in output however will be temporary and output will be back at the LRAS level in the long run.

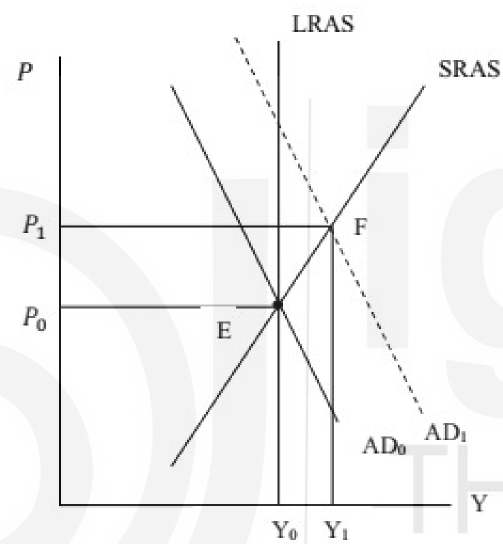

Fig. 3.9: Impact of Demand Shock

Check Your Progress 3

1) Explain how the AD curve is derived?

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2) Point out the factors that influence the position and slope of the AD curve.

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3.6 LET US SUM UP

In this Unit we described the IS-LM model, which signify the equilibrium in the real sector and the monetary sector of the economy. We derived the IS curve – on each and every point of the IS curve the real sector is in equilibrium. On the left side of the IS curve there is excess demand for goods while on the right side there is excess supply. The LM curve is the combination of all such points where the monetary sector of the economy is in equilibrium. On the right side of the LM curve, there is excess demand for money while on the left side there is excess supply of money. If the economy is operating at a point away from the IS and LM curves, it has a tendency to move towards the equilibrium point.

We derived the AD curve on the basis of IS-LM model. The AD curve traces the combinations of equilibrium output and prices. The equilibrium output and prices in the economy derived from the AS-AD model is also discussed.

3.7 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) It will affect both the intercept and slope of the IS curve. Look into equation (3.4). Decrease in the marginal propensity to consume (coefficient 'b') will lead to a decrease in the value of the intercept term. Look into the last term of the equation. It will increase the value of $\frac{e}{(1-b-d)}$. Thus, the IS curve will be steeper.

2) (i) Decrease in autonomous government expenditure will lead to a parallel downward shift in the IS curve to the left.

   (ii) Decrease in lumpsum tax will lead to a parallel upward shift in the IS curve. But the question is about decrease in tax rate, which depends on income (Y). Thus, it will affect the slope of the IS curve. The IS curve will become flatter (why? You can take a tax function $T = t_0 + t_1 Y$, substitute it in place of T in equation (3.4) and find out).

Check Your Progress 2

1) The LM curve is derived from the equality between the demand for and supply of real balances. See Section 3.3, particularly Fig. 3.3.

2) When the LM curve is vertical, demand for money is insensitive to interest rate. This is a case of liquidity trap situation.

3) Go through Sub-Section 3.4.3 and answer.

Check Your Progress 3

1) The AD curve is derived from the IS-LM model. Go through Sub-Section 3.5.1 and explain Fig. 3.8.

2) Since the AD curve is derived from the IS-LM model, it depends on the factors that influence the IS and LM curves. Explain the impact of these factors on the AD curve.

UNIT 4 OPEN ECONOMY MACROECONOMICS -I*

Structure

4.0 OBJECTIVES

After going through this Unit you should be in a position to

- explain the national income identities in an open economy;
- assess the relation between aggregate expenditure and income in an open economy;
- explain the relation between private savings-investment gap, fiscal deficit and current account deficit;
- explain the concept of Balance of Payments (BOP);
- explain current and Capital accounts of BOP;
- explain the concepts of nominal and real exchange rates; and
- explain fixed and flexible exchange rate regimes.

4.1 INTRODUCTION

In this Unit we will examine the national income identities, with a view to identifying certain fundamental differences between open and closed economies. Thereafter we examine certain basic concepts relating to Balance of Payments (BOP) and exchange rates. The BOP accounts record the transactions of an economy with the 'rest of the world' and provide important insights into the pattern of international trade and capital flows. The exchange rate is also an important variable in an open economy with important implications for international trade and competitiveness.

4.1 NATIONAL INCOME IDENTITY IN AN OPEN ECONOMY

In an open economy domestic agents have manifold economic interactions with foreigners; e.g., goods are sold to, as well as purchased from foreigners. The sale of goods and services to foreigners are exports (X) of the country and purchases from foreigners constitute imports (M). In an open economy, aggregate expenditure on final goods by domestic agents (households, firms and

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government) given by [C + I + G], includes expenditure on domestic as well as imported goods. Aggregate spending on *domestic* goods by domestic agents is given by [C + I + G – M]. To this we add expenditure on domestic goods by foreigners or exports, to obtain *total* final expenditure on domestically produced goods, i.e., [C + I + G + X –M].

From the equivalence between the expenditure, income and product approaches to measurement of aggregate output, we can say that aggregate final expenditure is equal to aggregate output produced in the economy or GDP (Y).

So the national income identity for an open economy can be written as:

$Y = C + I + G + X – M$

or, $Y = C + I + G + NX$ (where $NX = X – M$) (4.1)

Note that (X – M) is also referred to as net exports (NX) or balance of trade. When exports exceed imports, NX is positive and there is a trade surplus; when imports exceed exports, NX is negative and there is a trade deficit.

In an open economy we draw a distinction between GDP and GNP (Gross National Product). While GDP refers to aggregate incomes earned (by domestic residents and foreigners) within the geographical boundaries of the country, GNP is defined as the aggregate income of citizens of the country, irrespective of whether they are located within the country or abroad. E.g., the salary of a German consultant located in New Delhi would be a part of German GNP and India's GDP; the salary of an Indian worker in Singapore would be a part of Singapore's GDP and India's GNP.

So we define net factor income from abroad (NFIA) as follows:

NFIA = Income earned abroad by domestic factors of production *minus* income earned by foreign factors of production in the domestic economy.

To obtain GNP, NFIA is added to GDP (denoted often as Y in macroeconomics):

i.e., $GNP = Y + NFIA$ (4.2)

Note that NFIA can be positive or negative and the difference between GDP and GNP can often bequite large.For example, GDP may exceed GNP (NFIA < 0) in countries where mostproduction units are owned by foreign multinationals (so that a large share of the incomes generated within the economy, accrue to foreigners) or which have a sizeable migrant population employed abroad.

Using the national income identity we can show in an open economy aggregate expenditure (C+I+G) can exceedaggregate income (GNP) and that this imbalance is reflected in a current account deficit.

We can also write (4.2) as:

$GNP = C+I+G+NX +NFIA$

or, $GNP = C+I+G+ CA$, (where $CA = NX + NFIA$)

or, $GNP – (C+I+G) = CA$ (4.3)

Note that CA represents the 'current account balance' of a country, which may be positive (current account surplus) or negative (current account deficit).

The identity (4.3) indicates that aggregate domestic absorption (C+I+G) exceeds aggregate income (GNP) in a country that has a current account deficit ($CA < 0$). Such a country is a net debtor (or, borrower) as it has to borrow to bridge the gap between income and expenditure whereas aggregate income exceeds expenditure in a country running a current account surplus ($CA > 0$) and it is a net lender.

A current account deficit may be financed by borrowing from the rest of the world or by sale of foreign assets held by domestic agents. We will have more to say on this in the next section on balance of payments.

The national income identity can be further used to understand whether the real sector imbalance reflected in a current account deficit, stems mainly from the private sector, or the government sector, or both.

From the income side, we know that aggregate income is either consumed (C), or saved (S) or paid out in taxes (T), so that,

$$C + S + T = GNP \tag{4.4}$$

Combining (4.3) and (4.4), we can write,

$$C + S + T = GNP = C+I+G+CA$$

or, $C + S + T = C+I+G+CA$,

or, $S + T = I+G+CA$

$$\text{or, } (S - I) + (T - G) = CA \tag{4.5}$$

From identity (4.5) it is clear that a current account deficit ($CA < 0$) reflects either an excess of private investments over private savings (i.e., $(S - I) < 0$) or, a budget deficit (i.e., $(T–G) < 0$) or, both. Also, if $(S - I) > 0$ but, $(T–G) < 0$ and absolute value of (T–G) is larger than that of (S – I), there would be a current account deficit. The identity (4.5) represents the 'twin deficits', where an external deficit ($CA < 0$) is reflected as a deficit in the private sector (i.e., $(S - I) < 0$), or in the government sector (i.e., $(T–G) < 0$), or in both.

The national income identities clearly indicate that an imbalance in the current account reflects an imbalance in the real economy.That is, whenever a country runs a current account deficit, there is an excess of aggregate expenditure over income; this may occur because, private savings are too low, or there is aprivate investment boom, or, the government is running a budget deficit or some combination of these factors. Conversely, aggregate income exceeds expenditure in a country that runs a current account surplus and either its private savings would exceed investments (i.e. $(S - I) > 0$) or, public savings would be positive (i.e., $(T - G) > 0$) or, both.

The main difference between closed and open economies should be evident from the above discussion. In a closed economy, by definition, there are no transactions with the 'rest of the world'and CA is zero. Thus aggregate expenditure cannot exceed aggregate income. It means, an excess of private investments over savings ($S - I < 0$) needs to be compensated by public savings ($T–G > 0$). However, open economies can borrow from the rest of the world. Thus it can sustain excess of expenditure over income and excess of investments (I) over private (S) and public savings ($T - G$). International trade and capital flows allow countries running current account deficits to draw on the savings of other countries running current account surpluses.

Check Your Progress 1

1) Use the national income identity for an open economy and show that in a country running a trade deficit (i.e., $NX < 0$): (a) aggregate expenditure must be greater than GDP, and (b) national savings (i.e., sum of private savings and public savings) must be less than investments.

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2) Use the national income identity to bring out at least two fundamental differences between open and closed economies.

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4.3 BALANCE OF PAYMENTS AND EXCHANGE RATE

The Balance of Payments (BOP) of a country is a record of all its transactions with foreigners during a given period (typically one year). The transactions include those involving purchase and sale of goods, services, physical and financial assets, as well as transfer payments.

4.3.1 Balance of Payments

The BOP account has two main components, viz., the current account and the capital account.

Current Account: The current account of BOP records receipts from and payments to foreigners due to international trade in goods and services (including factor services). In the current account transactions involving *inflows* of foreign exchange (e.g. exports) are a credit and those involving *outflows* of foreign exchange (e.g., imports) are a debit entry. The current account comprises two main components: the balance of trade and the balance on invisibles. Transactions involving trade in services, factor incomes and transfers are also referred to as 'invisibles' in the current account. These are further explained below.

The balance of trade or merchandise trade balance is the difference between exports and imports of goods by a country and it may be positive, negative or zero. A negative balance of trade or a trade deficit indicates that the country's imports of goods exceed exports; a positive trade balance indicates the country has a trade surplus, with exports of goods exceeding imports.

Trade in services (e.g., India's software exports) is a part of invisibles in the current account. The balance of trade in services captures the difference between a country's service exports and imports and it may also be positive or negative, depending on whether the country is a net exporter or net importer of services.

Factor incomes include *net investment incomefrom abroad* which measures investment earnings from abroad earned by Indians *minus* foreigners' earnings from their Indian assets. Indian residents may receive investment income, such as dividends and interest, on the foreign assets they hold while foreign residents receive investment income on the Indian assets they possess. The former is a credit (inflow of foreign exchange) while the latter is a debit entry (outflow of foreign exchange) in the current account.

Unilateral Transfers, that are overseas payments made without any *quid pro quo*, are also recorded in the current account. These include assistance by foreign governments during war or natural calamity (foreign aid), workers' remittances (e.g., money sent by overseas Indians to their families in India), personal gifts, donations etc. The *net unilateral transfers* are transfers received by Indians from abroad minus similar transfers sent to foreigners from India. For many developing countries workers' remittances are an important source of foreign exchange earnings.

The current account balance is the sum of (i) the balance on merchandise trade and (ii) the balance on invisibles (which comprises balance on services trade, net investment income and net transfers). When total foreign exchange receipts on account of all the transactions recorded in the current account are greater than total payments, the country has a current account surplus. In case the foreign exchange payments exceed the receipts, the country has a current account deficit.

Note that even if a country has a trade deficit it could have a current account surplus, if it has a positive and very high net invisibles balance. This can be seen in many countries that are net recipients of large unilateral transfers and remittances. For example, remittances received from Indian workers employed abroad are an important source of foreign exchange earnings for India. India is the largest recipient of remittances in absolute figure ($87 billion in 2021) while in terms of percentage of GDP Lebanon is the largest recipient that contributed 54 per cent of GDP in 2021.

Capital Account: The capital account of the BOP includes transactions involving cross-border purchase and sale of real and financial assets. The asset in question could be physical assets like land, factories, houses or financial assets like stocks and bonds. In the capital account each transaction is recorded twice, once as a credit and once as a debit entry, thereby capturing the two aspects of each transaction. E.g., purchase of a foreign financial asset such as foreign government bond or shares of a foreign firm by a domestic resident involves an asset import and a capital outflow (payment made by the domestic resident for the foreign asset).

Broadly two types of capital flows are recorded in the capital account – debt creating and non-debt creating flows. For example, when domestic agents borrow from foreign financial institutions, the country experiences a debt creating capital inflow, as this loan has to be repaid at a future date. However, foreign firms investing in the domestic country (e.g., foreign direct investment or FDI), involve a non-debt creating capital inflow, as this does not create any commitment for repayment for the recipient nation. Capital flows can also be classified on the basis of the duration involved. For example, foreign institutional investors (FIIs) purchasing shares of domestic firms, involves a short term capital inflow (also known as foreign portfolio investment or FPI). The FIIs may sell the equity and withdraw their funds at a short notice. Such withdrawal of foreign funds would involve a capital outflow. Typically, investment by foreign firms in country's production sector (e.g., Greenfield FDI) involves longer term capital inflows that are not likely to be withdrawn at short notice unlike investment in the financial sector (e.g., FPI).

When capital inflows exceed capital outflows, the country has a capital account surplus. In the opposite case, when outflows exceed inflows it has a capital account deficit.

In the current account, in general, transactions involving the inflow of foreign exchange are recorded as a credit entry, with the corresponding debit entry appearing in the capital account. Therefore a surplus (or deficit) in current account is mirrored in a corresponding deficit (or surplus) in the capital account. If a country has a current account deficit, this must be financed either by selling assets or by borrowing, so there must be a corresponding surplus in the capital account. So a current account deficit is financed by net capital inflows, either on account of the private sector or the government sector or both.

Overall BOP Balance: Every transaction is recorded twice in the BOP account, once as a debit and once as credit, as per the principles of double entry bookkeeping. Therefore in an accounting sense the BOP is always balanced. However, in practice, this is not observed, primarily because of problems related to data collection. There are time lags involved in the international transactions; e.g., data on exports is collected from customs, at the time goods are shipped, whereas payments for the same may be received six months later and hence may not be recorded in BOP of that year. Therefore, in order to balance the BOP account an entry for *errors and omissions* is included.

Even though the BOP always balances, we can talk about an overall BOP surplus or deficit. Overall BOP balance is obtained by adding the current and capital account balances. A country has a BOP surplus in the following cases: (i) there is current account surplus as well as capital account surplus; (ii) there is a current account deficit but a capital account surplus that is larger in magnitude; (iii) there is a current account surplus and a relatively smaller capital account deficit. There would be a BOP deficit, either when there is a deficit on both current and capital accounts or when there is a relatively large deficit in either one of the current or capital accounts.

In the case of a BOP surplus there is an increase in the foreign exchange reserves of the country. In the case of a deficit, the stock of foreign exchange reserves is depleted. Let us discuss this issue in details.

Change in Foreign Exchange Reserves

Apart from transactions made by private and government agents, the capital account of the BOP also includes *official reserve transactions* that involve change in the stock of foreign exchange reserves held by the central bank. When there is a BOP surplus, the country experiences a net inflow of foreign exchange, leading to a correspondingincrease in foreign exchange reserves. In the case of a BOP deficit, there is net outflow of foreign exchange and a corresponding reduction in foreign exchange reserves. Thus,

BOP Balance = Current Account + Capital Account (including errors and omissions) + Official Reserve Transactions = 0

In case a country has a current account deficit (of say, – $15,000), but a capital account surplus that is smaller in magnitude (say, $12,000), then it has a BOP deficit (of –$3000) that leads to a decrease in foreign exchange reserves (by $3000).

Note that a BOP surplus is recorded with a positive sign in the BOP, while the corresponding *increase* in reserves has a *negative* sign, so that the two add up to zero. Therefore in BOP accounts an increase in foreign exchange reserves appears with a negative sign whereas a decrease in reserves has a positive sign.

4.3.2 Exchange Rate

The exchange rate is an important concept in an open economysuch as India, that has transactions involving foreign goods, services and assets, whose prices are denominated in terms of foreign currencies. Throughout the following discussion we assume that the domestic country is India and the foreign country is the USA.

Demand for foreign exchange arises whenever domestic agents purchase foreign goods, services or assets (i.e., imports and capital outflows); whereas, sale of domestic goods, services and assets to foreigners (i.e., exports and capital inflows) constitutes the primary source of supply of foreign exchange. The market value of the exchange rate is determined by demand and supply in the foreign exchange market. However, often the official exchange rate may differ from its market value as you will see from our discussion on alternate exchange rate regimes. First we shall learn about various exchange rate concepts.

Nominal Exchange Rates

The nominal exchange rate 'e' can be defined as the number of units of domestic currency (Rupee) required forpurchasing one unit of foreign currency (dollar). For example, if Rs. 80 is required to buy one dollar, then the exchange rate e = Rs. 80 per dollar. Similarly, the exchange rate of the Indian rupee can also be defined in terms of other currencies, e.g., yen, euro, pound, etc.

When the exchange rate 'e' rises (say, from Rs. 80 to Rs.82 per dollar), there is an increase in the price of dollars in terms of rupees and there is ***nominal depreciation*** of the rupee vis-à-vis dollar (and ***nominal appreciation*** of the dollar vis-à-vis rupee). When 'e' falls (e.g., from Rs. 80 to Rs. 78 per dollar), the dollar becomes cheaper in terms of rupees and there is a ***nominal appreciation*** of the rupee (and ***nominal depreciation*** of the dollar vis-à-vis rupee).

Real Exchange Rate

While nominal exchange rates measure the rate of exchange between domestic and foreign currencies, the real exchange rate captures the rate at which domestic goods are exchanged for foreign goods. It isthe price of foreign goods in terms of domestic goods and is computed using the nominal exchange rate and prices of a comparable basket of domestic and foreign goods as bellow:

$$r = (e\, p_f) / p_d \qquad (4.6)$$

where'e' is the nominal exchange rate (rupees per dollar), p_d denotes the domestic price level (in rupees) and $p_f$denotes the foreign price level (in dollars). 'e p_f' is the price in rupees of a foreign basket of goods and 'p_d' is the price (in Rupees) of a comparable domestic basket.

The real exchange rate between rupee and dollar reflects the price of goods produced in the USA relative to those produced in India. A rise in r means American goods become relatively more expensive vis a vis Indian goods and there is ***real depreciation*** of the rupee; whereas a reduction in r makes Indian goods relatively more expensive and there is***real appreciation*** of the rupee. The real exchange rate is often used as an index of a country's international price competitiveness. In case a country faces real appreciation of its currency, domestic goods become more expensive and this can lead to a reduction in demand for exports. In contrast, a real depreciation of the currency can boost export demand.

When domestic and foreign prices remain unchanged (that is, $\frac{p_f}{p_d}$ is constant), a nominal depreciation ('e' rises) of the domestic currency causes a corresponding real depreciation ('r' rises). In this case the nominal and real exchange rates move in the same direction. However, prices may change simultaneously with exchange rates; e.g., an appreciation in nominal exchange rates ('e' falls), along with a large fall in domestic prices (p_d falls) may lead to a real depreciation (r rises). So, 'e' and 'r' may also move in opposite directions.

Exchange Rate Regimes

The exchange rate regime of a country determines how its exchange rate is determined.

Under a ***flexible (or floating) exchange rate regime***, the value of 'e' is determined by the demand and supply of foreign exchange, without any government intervention. In this case the currency is on a clean float or pure float. That means the exchange rate is fully flexible and adjusts continuously to equate market demand and supply of foreign exchange, so there is no role for intervention by the central bank and hence no need of foreign exchange reserves. In this case BOP balance is attained via exchange rate fluctuations. That is, in case there is a BOP surplus (supply of foreign currency exceeds demand), the exchange rate appreciates (e falls, so that given prices, r falls, so foreign goods are relatively cheaper), X falls, M rises and NX falls (the current account deficit widens and BOP surplus is reduced), till BOP equilibrium is restored. Floating exchange rates lead to considerable volatility in exchange rates that may be unfavorable for exporters and importers and may have adverse impacts on domestic production, investment and employment. As such, 'pure floats' are rarely observed in practice.

In most countries central banks maintain a stock of foreign exchange reserves and intervene and try to maintain the value of exchange rate within a certain band. Such exchange rate arrangements, widely prevalent across countries, are called a 'managed float' or dirty float. ***Intervention*** refers to the central bank buying or selling foreign exchange in an attempt to influence the exchange rate. In a managed float, the currency is allowed to float within a certain 'target zone' and the central bank stands ready to intervene (e.g., buy or sell dollars against rupees) whenever the exchange rate appears to breach the limits of this band.

At the other extreme we have the system of fixed exchange rate, wherein the exchange rate is not freely floating, but is fixed by the government. An important function of central banks in a fixed exchange rate system is to intervene in foreign exchange markets to keep the price of foreign exchange fixed at the pre-announced level. For this, central banks have to maintain adequate stock of foreign exchange reserves.

Under a fixed exchange rate system, the fixed parity announced by the central bank may be revised at times; in this case a decrease in 'e' due to official intervention is called ***revaluation*** while an increase in 'e' by official intervention is called ***devaluation***. In a world with few restrictions on international transactions in goods and assets, maintaining a fixed exchange rate poses an immense challenge for a country's government. We will discuss a few aspects of policy dilemmas under a fixed exchange rate regime in the next section.

Check Your Progress 2

1) State whether the following statements are 'True' or 'False', giving reasons for your answer in each case:

   (a) A country can have a current account surplus, even when it has a negative trade balance (or merchandise trade deficit).

   (b) When Indian firms invest in foreign firms ('outward FDI'), there would be a corresponding capital inflow in the current account of India's BOP.

   (c) A country can have an overall BOP deficit even when it has a current account surplus.

   (d) When a country has a current account deficit (of say –\$20,000) and a capital account surplus (of say \$8000), its foreign exchange reserves would increase.

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2) Suppose that domestic currency depreciates by 2 percent and simultaneously domestic prices rise by 5 percent, while foreign prices remain unchanged.

   (i) What would be the impact on (a) real exchange rate; and (b) net exports?

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   (ii) How would your answer in (b) above wouldchange if domestic prices remained constant following the nominal currency depreciation?

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3) Suppose a country experiences a sudden surge in capital inflows (e.g., FPI inflows) that create an excess supply of foreign exchange (say dollars). How would this affect the following?

(a) the exchange rate and net exports under a flexible exchange rate regime

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(b) the exchange rate and foreign exchange reserves under a fixed exchange rate regime ?

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4.4 LET US SUM UP

In this unit, we learnt that how openness to international trade and capital flows affects policy and macroeconomic outcomes within an economy. Our discussion on national income identities highlighted important differences between closed and open economies. For example, we saw that open economies can run current account deficits and borrow from the rest of the world to cover an excess of aggregate expenditure over income. Exchange rates are an important macroeconomic variable in open economies and from our analysis of BOP we got a flavor of the various factors, related to international trade in goods, services and assets that affect the demand and supply of foreign exchange and have an impact on nominal and real exchange rates. In this context we also learnt about alternate exchange rate regimes, viz., fixed and flexible exchange rate regimes.

4.5 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) Refer to Section 4.2 and answer.

2) Refer to Section 4.2 and answer

Check Your Progress 2

1) a) True

b) False

c) True

d) False

2) Refer to Sub-section 4.3.2 and answer.

3) Refer to Sub-section 4.3.2 and answer.

UNIT 5 OPEN ECONOMY MACROECONOMICS -II*

Structure

5.0 OBJECTIVES

After going through this Unit you should be in a position to

- analyse short run macroeconomic equilibrium using the open economy IS-LM model; and
- discuss the limits to monetary and fiscal policies under fixed vis-à-vis flexible exchange rate regimes.

5.1 INTRODUCTION

In this Unit we will introduce the open economy IS-LM framework and use it to understand the complications involved in framing monetary and fiscal policy under alternate exchange rate regimes. In particular we consider two different exchange rate regimes, viz., fixed and flexible rate systems and explore the implications for conduct of monetary and fiscal policies under the assumption of unrestricted international capital mobility.

Apart from the above, we use the open economy IS-LM framework with perfect capital mobility to study the conduct of monetary and fiscal policy under alternate exchange rate regimes. First we examine the conduct of policy making under flexible exchange rates, followed by the case of a fixed exchange rate regime. This is known as the 'Mundell-Fleming model', named after economists Robert Mundell and Marcus Fleming, who developed this framework and analysed policymaking in open economies in the early 1960s.

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5.2 THE OPEN ECONOMY IS-LM FRAMEWORK

After our discussion above on certain basic concepts and accounting tools pertaining to open economies (see Unit 4), we will now examine output market equilibrium in the short run Keynesian framework using the open economy IS-LM model.

As in case of the closed economy IS-LM model, here also we assume that prices are given and there is 'excess capacity', so that output is demand determined.

5.2.1 The IS Curve

You are familiar with the IS curve in a closed economy (see Unit 3). In an open economy, aggregate demand (AD) is given by:

$$AD = C + I + G + X - M = C + I + G + NX, \qquad (5.1)$$

where, $C = C(Y_d)$, is aggregate consumption expenditure which is a function of disposable income, $Y_d = Y - T$;

$I [= I(i)]$ is private investments, an inverse function of the rate of interest i;

G represents exogenous government expenditure;

$X [= X(Y_f, r)]$ is exports, the demand for domestic goods by foreigners, which is a function of foreign GDP (Y_f) and the real exchange rate r;

$M [= M(Y, r)[$ is imports, the demand for foreign goods by domestic agents, which is a function of domestic GDP (Y) and the real exchange rate r;

$NX [= (X - M) = NX(Y_f, Y, r)]$ stands for the net exports that depend on the factors underlying demand for imports and exports, i.e., Y_f, Y and r.

Clearly imports are a leakage from the domestic economy as it is the expenditure ofdomestic agents on output that is produced in a different economy (hence part of their GDP) while exports are an addition to the total expenditure on domestic goods. Exports depend on Y_f, *ceteris paribus*, the higher is Y_f, higher would be export demand. Similarly, the higher is Y, higher would be the demand for imports. Both exports and imports depend on the real exchange rate which represents the relative price of foreign vis-à-vis domestic goods as discussed in equation (6). With an increase in r (a real depreciation), foreign goods become relatively more expensive. Thus, there is a decline in imports, while exports increase as domestic goods become relatively cheaper. So a rise in r would lead to a rise in X and a fall in M and net exports would be higher. Note that economists Alfred Marshall and Abba Lerner gave certain conditions under which a real devaluation or depreciation leads to an improvement in the trade balance (increase in NX), known as the *Marshall- Lerner condition*. According to the Marshall-Lerner condition, depreciation will lead to an improvement in the balance of trade of a country only if the sum of the price-elasticities of exports and imports of the country is greater than one. Throughout our discussion we assume that the Marshall-Lerner condition is satisfied.

Note that one of the assumptions in the IS-LM framework is that of given prices. Therefore, with given foreign (p^f) and domestic prices (p^d), a nominal depreciation or appreciation (rise or fall in e) brings about a real depreciation or appreciation (rise or fall in r).

As in the closed economy case, the IS curve, represents goods market equilibrium, so that along the IS curve we have aggregate demand equal to aggregate output or GDP (Y):

i.e., Y = AD

or, $Y = C(Y_d) + I(i) + G + NX(Y_f, Y, r)$ (5.2)

However, compared to the closed economy case, now there are two new determinants of aggregate demand, viz., foreign GDP and the real exchange rate. These two factors affect net exports and hence demand and output in an open economy. If Y_f increases (decreases) or r increases (decreases), NX would also increase (decrease) and the IS curve would shift to the right (left).

Another aspect of open economies reflected in (5.2) is that any change in aggregate demand (e.g., due to an increase in G), would affect Y and hence M and therefore lead to a change in NX. For instance, *ceteris paribus* an increase in government spending would lead to an increase in Y and M and thus reduce NX (i.e. reduce the size of the trade surplus or widen an existing trade deficit). However, if foreign GDP Y_f increases, *ceteris paribus*, exports would increase (as these are imports of the foreign country), also leading to an increase in Y, but in this case the trade balance would improve, i.e., NX would rise (assuming that rise in X due to increase in Y_f, would be more than the rise in M induced by rise in Y).

5.2.2 The LM Curve

So far as the LM curve is concerned, openness of an economy does not fundamentally change the inverse relation between money demand and the rate of interest (see Unit 3). If the demand for money is higher, the lower is the rate of interest, as in the case of a closed economy. Thus the LM curve, representing money market equilibrium (equality between demand for money and supply of money) is upward sloping. However, in an open economy, asset transactions with the rest of the world would give rise to capital flows (recorded in the capital account of the BOP). This also affects the rate of interest and money supply in an open economy. We discuss further on this issue below.

Let us assume that there is ***perfect capital mobility***, i.e., there are no restrictions on international capital inflows and outflows and cross-border capital movements involve zero transaction costs. This is a simplifying assumption that captures the reality of large-scale international capital mobility observed in the era of globalization, with advances in technology greatly reducing the cost of moving capital across international borders.

The assumption of perfect capital mobility has certain implications for the determination of interest rates in an open economy. It means that the domestic and foreign bonds are perfect substitutes, so that therate of intereston domestic bonds (i)has to be in line withthe rate of interest on foreign bonds (i_f), i.e., $i = i_f$ must hold. Here we assume that the foreign interest rate is given and not influenced by domestic economic policy.

If i_f is higher than i (i.e., $i_f > i$), foreign bondswould be preferred by investors owing to relatively higher returns on them. In this case, the domestic economy would experience capital outflows, as agents sell domestic bonds and buy dollars to purchase foreign bonds. The outflows continue until i rises (as price of domestic bonds fall) and equality is restored between i and i_f. When $i = i_f$, economic agents (such as households and firms) are indifferent between domestic bonds and foreign bonds.

Another point you should note regarding the money market in open economies relating to the supply of money and the role of foreign exchange reserves. Note that the stock of foreign exchange reserves is an asset of the central bank. When the central bank buys foreign currency, adding to its assets, it issues domestic currency so there is matching rise in its liability. As such, any change in foreign exchange reserves affects the monetary base or high powered money (liabilities of the central bank) and hence money supply in the economy, via the money multiplier.

Therefore, when a country experiences capital inflows and has a BOP surplus, this increasesforeign exchange reserves and leads to an increase in money supply. Similarly, a capital outflow can result in a BOP deficit and a fall in money supply via a reduction in the stock of foreign exchange reserves.

In this context you should note that the central bank often uses a policy of ***sterilization*** to offset the impact of a change in foreign exchange reserves on the money supply. This can be done by using open market operations. In particular, central banks often use foreign exchange reserves for intervention in the foreign exchange market. Suppose the RBI intervenes to buy dollars (in order to absorb an excess supply of dollars) against rupees. This would add to its foreign exchange reserves and hence increase money supply. However, the RBI can carry out a sterilized intervention, by simultaneously selling bonds, in an open market operation, to reduce money supply. Thereby the RBI can either fully or partially offset the impact of the increase in foreign exchange reserves on the money supply.

5.2.3 Equilibrium in Open Economy IS-LM Model

As in case of a closed economy, macroeconomic equilibrium occurs at the intersection of the IS and LM curves as shown in Fig. 5.1.

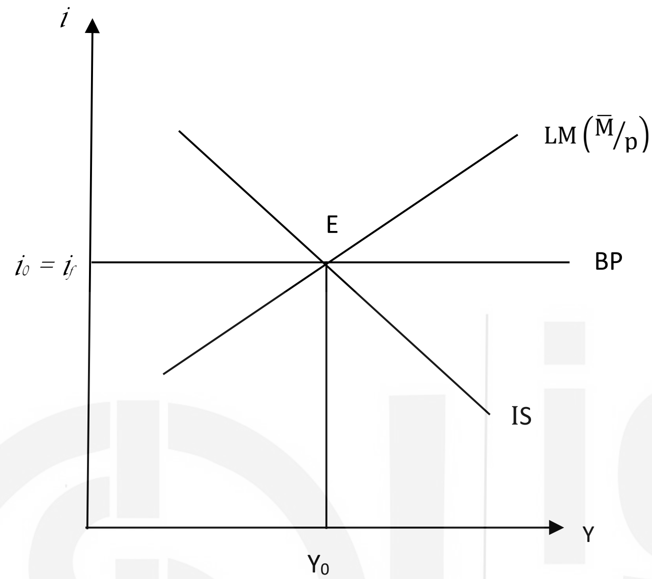


Fig. 5.1: "Macroeconomic Equilibrium in Open Economy IS-LM Model

Note that at the equilibrium E,the rate of interest, $i_0 = i_f$, owing to the assumption of perfect capital mobility. The point E represents macroeconomic equilibrium in the sense that there is both internal balance (goods and money market equilibrium) and external balance (BOP equilibrium).

The horizontal line BP that passes through the equilibrium rate of interest represents BOP equilibrium, in the sense that it represents interest rate – income combinations for which the current account surplus (deficit) is exactly offset by a capital account deficit (surplus). Note that here NX represents the current account balance.

With perfect capital mobility, the BP curve is horizontal, as the slightest deviation of i from i_f, would create BOP disequilibrium and trigger either infinite capital inflows or outflows. If $i > i_f$, there will be a surge in capital inflows and a BOP surplus, since inflows would be higher than required for BOP balance, at any given level of income. Whereas, or any $i < i_f$, there will be unlimited capital outflows and a BOP deficit. That is, for any Y (say Y_0), there is BOP surplus at all points above BP and a BOP deficit at all points below BP.

With imperfect capital mobility the BP curve would be upward sloping. As Y increases, M would rise, NX would fall and the current account deficit would widen. To maintain BOP balance therefore, i would have to increase to attract capital inflows and create a matching capital account surplus. With imperfect capital mobility, which occurs when there are restrictions on cross-border capital flows or when domestic and foreign bonds are imperfect substitutes, there can be a difference between iand i_f. However, here we will only consider the case of perfect capital mobility.

5.3 MONETARY AND FISCAL POLICIES UNDER FLEXIBLE EXCHANGE RATE

When a country has a flexible exchange rate regime, its exchange rate is determined by demand and supply in the foreign exchange market. We consider the case of a *pure float*, where there is no official intervention in currency markets and hence no role of foreign exchange reserves, with BOP equilibrium achieved via exchange rate fluctuations.

5.3.1 Fiscal Policy under Flexible Exchange Rates

Consider the open economy IS-LM model with perfect capital mobility and suppose the government follows expansionary fiscal policy in this set up as in Fig. 5.2.

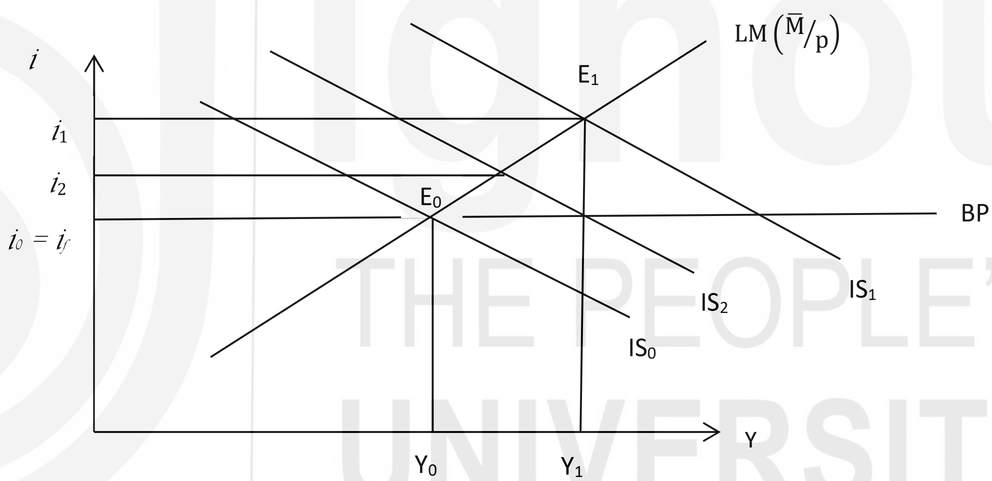
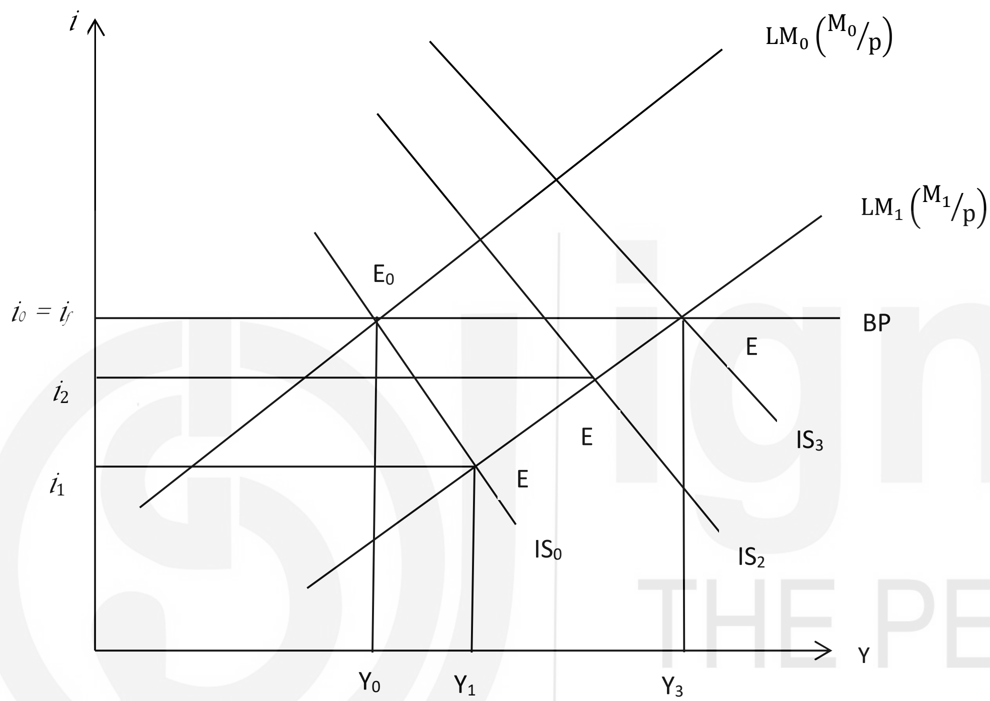


Fig. 5.2: Expansionary Fiscal Policy under Flexible Exchange Rates

Initially the economy is at E_1 (see Fig. 5.2), where goods and money markets are in equilibrium and there is BOP equilibrium as well, with $i_0= i_f$. With fiscal expansion (or fall in tax rate t), there is an increase in G. The IS curve shifts from IS_0 to IS_1 and the interest rises from i_0 to i_1 as the economy moves to the new equilibrium E_1. However, with i_f remaining the same, the rise in domestic interest rate makes domestic bonds relatively more attractive and there is a surge in capital inflows, that leads to a BOP surplus (excess supply of foreign exchange), causing a currency appreciation (e falls). With given prices (implying prices remaining constant), a nominal appreciation leads to a real appreciation (r falls), reducing X, raising M and worsening the trade balance (NX falls). As a result aggregate demand falls and the IS curve shifts to the left. It shifts back all the way to IS_0, where once again $i = i_0 = i_f$ and output is back to Y_0.Note that at any other IS curve, e.g., IS_2, the domestic interest rate is still above i_f ($i_2> i_f$), so e, r and NX would fall and IS would shift further to the left. Therefore, we see that with flexible exchange rates, fiscal expansion is ineffective. It fails to raise output (the increase in output from Y_0 to Y_1 is only temporary), the trade balance worsens because of appreciation in the exchange rate (NX is lower) and the fiscal position deteriorates (G has increased or tax rate t has fallen).

5.3.2 Monetary Policy under Flexible Exchange Rates

We now examine the impact of expansionary monetary policy under a flexible exchange rate regime.



[Fig. 5.3: Expansionary Monetary Policy under Flexible Exchange Rates]

In Fig. 5.3 an increase in nominal money supply (from M_0 to M_1) shifts the LM curve outward from LM_0 to LM_1 and the economy moves from E_0 to E_1, as domestic interest rate falls from i_0 to i_1.At i_1 there is internal balance (output and money markets are in equilibrium) but there is external imbalance (BOP disequilibrium) that triggers capital flows. Since $i_1 < i_f$, this triggers a capital outflow, leading to a BOP deficit (excess demand for dollars) and causing a depreciation of the nominal (e rises) and real exchange rates (r rises). As a result domestic goods become relatively cheaper, X rises, M falls and NX increases, and with higher external demand, the IS curve shifts to the right. This process continues till the IS curve shifts all the way to IS_3 and the economy is at E_3, where once again $i = i_f$ and both internal and external balance are restored. At any other point (e.g., E_2 on IS_2, $i_2 < i_f$, so e and r would rise, NX would fall, moving the IS curve further to the right).

This shows, that in complete contrast to fiscal policy, monetary expansion is an extremely effective policy tool in an open economy with flexible exchange rates and perfect capital mobility. It brings about currency depreciation that improves the trade balance, so that output increases from Y_0 all the way to Y_3.

Check Your Progress 1

1. What is the effect of monetary expansion on output and interest rates in an open economy with perfect capital mobility under flexible exchange rate?

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2. Comment on the effectiveness of monetary policy in open economies in light of your answer to Question 1 above.

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3. What is the effect of fiscal contraction (cut in tax rate) on output and interest rates in an open economy with perfect capital mobility under flexible exchange rates?

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5.4 MONETARY AND FISCAL POLICIES UNDER FIXED EXCHANGE RATE

With fixed exchange rates, exchange rate fluctuations are ruled out and the central bank plays a very important role as it holds foreign exchange reserves and intervenesin foreign exchange markets to maintain the fixed parity.

5.4.1 Fiscal Policy under Fixed Exchange Rates

We first consider the impact of expansionary fiscal policy under a fixed exchange rate system, where e = e* is the official parity (Fig. 5.4).

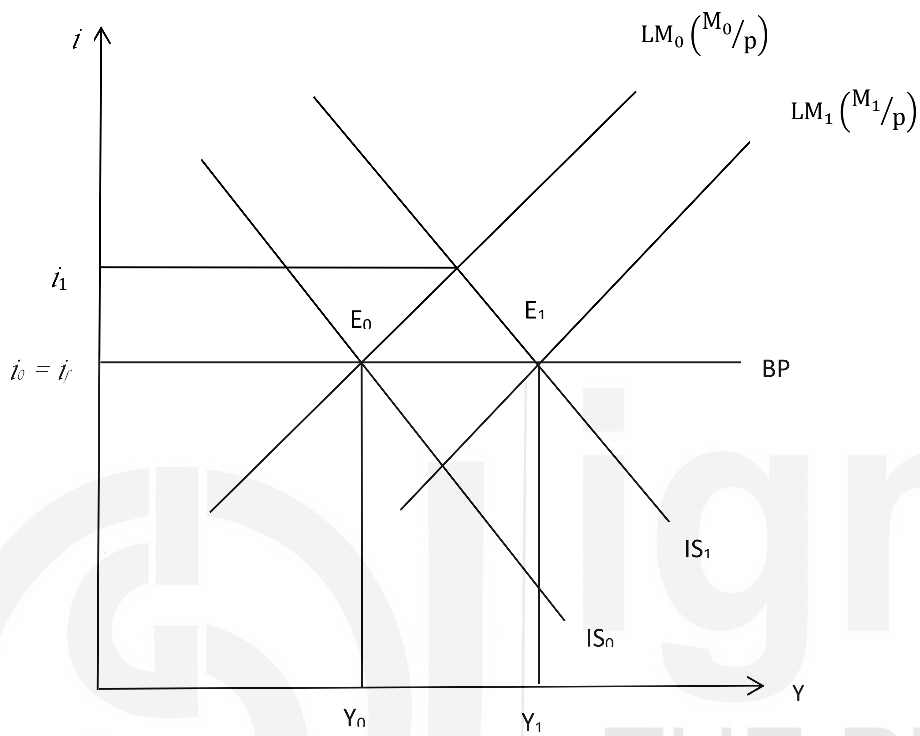

Fig. 5.4: Expansionary Fiscal Policy under Fixed Exchange Rates

Suppose there is fiscal expansion, either through an increase in government expenditure or via a cut in taxes. The IS curve shifts out from IS_0 to IS_1 and the interest rate rises to i_1 (see Fig. 5.4). With higher domestic interest rates ($i_1 > i_f$), there is a surge in capital inflows as there is BOP surplus and an excess supply of dollars (foreign exchange). This creates pressure for currency appreciation, but the central bank is committed to maintaining a fixed rate. Therefore it intervenes and buys dollars in order to mop up the excess supply. As a result foreign exchange reserves increase, leading to an increase in money supply, which shifts the LM curve outward. This process continues as long as $i > i_f$ and it stops only when LM shiftsto LM_1, where money supply has increased to M_1 ($M_1 > M_0$) and the economy reaches E_1 where $i = i_f$, output has increased to Y_1 and there is no crowding out.

Therefore, fiscal policy is extremely effective under a system of fixed exchange rates, as change in foreign exchange reserves brings about monetary expansion to accommodate the fiscal expansion. This is in complete contrast to the case of flexible exchange rates under which fiscal policy was quite ineffective.

5.4.2 Monetary Policy under Fixed Exchange Rates

Now we consider the case of a monetary expansion under a system of fixed exchange rates (Fig. 5.5).

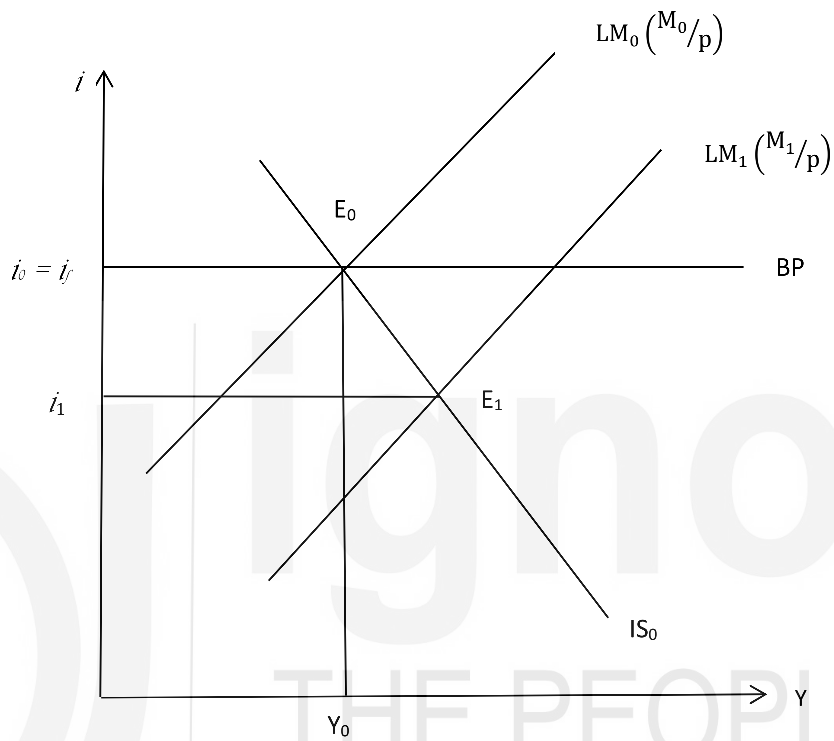

Fig. 5.5: Expansionary Monetary Policy under Fixed Exchange Rates

Suppose the central bank engages in an open market purchase of bonds in a bid to increase domestic money supply. This shifts the LM curve from LM_0 to LM_1(since money supply increases from M_0 to M_1) and the economy moves to E_1, where the interest rate $i_1 < i_f$ (see Fig. 5.5). Due to lower domestic rates of interest, investors would move to foreign bonds, triggering capital outflows and giving rise to a BOP deficit and excess demand for dollars. This creates pressure for currency depreciation, but the central bank is committed to maintain a fixed parity. Therefore it intervenes to sell dollars against rupees, leading to a fall in foreign exchange reserves and contraction in money supply. As money supply falls, the LM curve shifts to the left of LM_1 and this process continues as long as i< if. Ultimately the economy is back to E_0, with money supply falling back to its initial level (M_0), so that i = if and output is back to its initial level Y_0.

Clearly with fixed exchange rates and perfect capital mobility, money supply becomes ineffective as a policy tool as monetary expansion has no effect on output in the short run.

In fact the above discussion indicatesthat money supply becomes ***endogenous*** with fixed exchange rates and perfect capital mobility, in the sense that it passively adjusts to changes in the foreign exchange reserves as the central bank intervenes in foreign currency markets to defend the fixed parity. That is, since the central bank wants to hold the exchange rate constant, it has to intervene to keep the exchange rate from depreciating or appreciating, so foreign exchange reserves and hence money supply adjusts accordingly.

Our discussion also highlights the fact that it is impossible to simultaneously have the following three features in a single policy regime, viz.: (a) perfect capital mobility; (b) independent monetary policy; and (c) fixed exchange rates. This is also known as the '*policy trilemma*'. In the presence of perfect capital mobilityand *fixed* exchange rates, money supply becomes endogenous and the Central Bank's attempt to pursue an independent monetary policy isnot effectiveasthe Central Bank is committed to maintaining a fixed parity. Whereas, in the case of *flexible* exchange rates, perfect capital mobility and independent monetary policy do indeed go together, as we saw that monetary policy is very effective in this case. Countries may adopt fixed exchange rates and yet retain monetary policy independence only with restrictions on international capital flows, i.e., in the absence of perfect capital mobility. In this case, interest rate differentials between countries would persist owing to restrictions on capital mobility.

Check Your Progress 2

1. What is the effect of fiscal contraction (cut in tax rate) on output and interest rates in an open economy with perfect capital mobility under fixed exchange rate?

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2. Comment on the effectiveness of fiscal policy in open economies in light of your answer toquestion 1 above.

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3. (a) What is the effect of monetary contraction (sale of bonds by central bank) on output and interest rate in an open economy with perfect capital mobility under (i) fixed exchange rates, and (ii) flexible exchange rates?

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(b) Explain the concept of 'endogeneity' of money supply.

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(c) Comment on the effectiveness of monetary policy in open economies in light of your answer in part (a) above.

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4. Use the open economy IS-LM model to explain the concept of 'policy trilemma'.

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5.5 LET US SUM UP

In this Unit we discussed the open economy IS-LM model wherein we saw that international trade affects aggregate demand in an economy, while capital flows play an important role in shaping policy outcomes through their impact on interest rates, exchange rates and money supply. In particular we saw that with fixed exchange rates and perfect capital mobility, money supply becomes endogenous, i.e., countries cannot conduct independent monetary policy. Fiscal policy is an effective policy tool when the exchange rate is fixed; however, with flexible exchange rates and perfect capital mobility, fiscal expansion becomes ineffective as it has no effect on output and leads to a worsening of the trade balance. Finally, the effectiveness of monetary policy under flexible exchange rates demonstrates that countries can have at most any *two* of the three policy choices of having *independent monetary policy, fixed exchange rates* and *perfect capital mobility*.

5.6 ANSWER/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) Refer to Sub-section 5.3.1 and Fig. 5.3.

2) Refer to Sub-section 5.3.1.

3) Refer to Sub-sections 5.2.2 and 5.2.3 and Fig. 5.1

Check Your Progress 2

1) Refer to Sub-section 5.4.1 and Fig. 5.4.

2) Refer to Sub-section 5.4.1 and Fig. 5.4.

3) a) Refer to Sub-section 5.3.2 and 5.4.2, and Fig. 5.3 and Fig. 5.5.

   b) Refer to Sub-section 5.4.2, and Fig. 5.5.

   c) Refer to Sub-sections 5.3.2 and 5.4.2, and Fig. 5.3 and Fig. 5.5.

4) Refer to Sub-section 5.4.2.